Phishing Detection Through URL Structure

Group 2

MIS 430-01

Spring 2026

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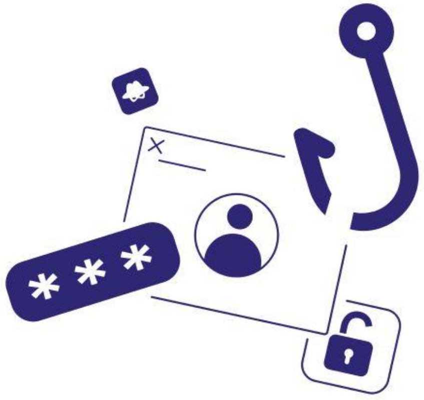
Introduction



"Phishing - is a type of cyberattack that uses fraudulent emails, text messages, phone calls or websites to trick people into sharing sensitive data, downloading malware or otherwise exposing themselves to cybercrime." (IBM, 2024).

Why it Matters?

1M+

Phishing attacks observed in Q1 2025 alone (APWG, 2025)

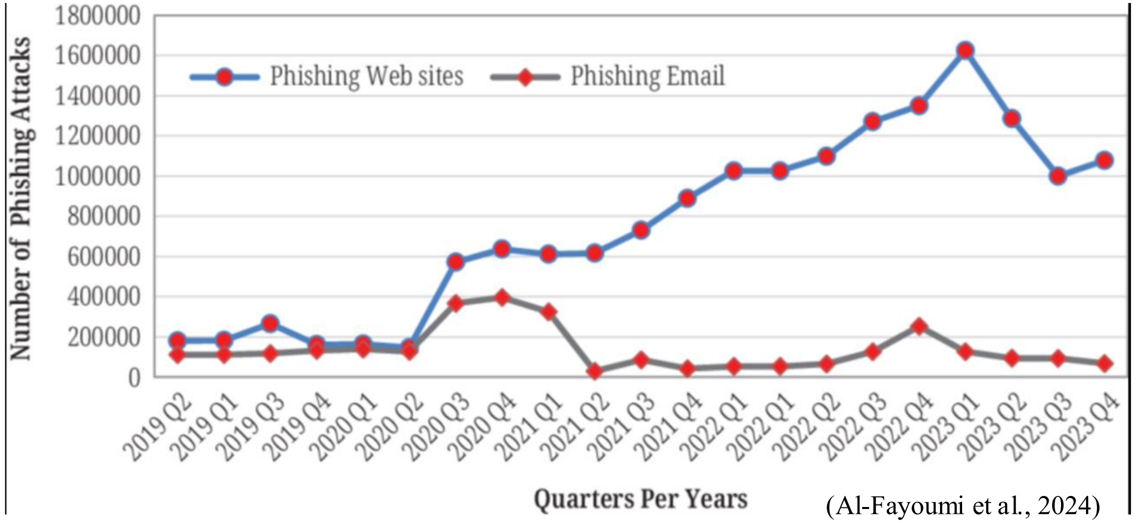
86%

Of organizations had at least one user click a phishing link (Zieni et al., 2023)

#1

Leading cause of cybersecurity breaches in 2023 (Verizon DBIR, 2023)

Phishing Statistics



(Al-Fayoumi et al., 2024)

Problem Identification

Attackers exploit subtle structural patterns in URLs: extra dots, misleading dashes, hidden numeric strings, and deceptive HTTPS wording.

Detecting these patterns at scale requires moving beyond intuition to data-driven analysis of a combination of URL features

Research Question

“Can URL structural features be used to predict Phishing attempts?”

Which features are the strongest predictors?

How effectively can URL features alone explain the complexity of phishing behavior?

Prior Work

- Mamun et al. (2016) – Lexical Analysis – identified dot count and share count as among the most influential URL features. Dashes are rarely used in legitimate domain.
- Kritika (2025) – Subdomain manipulation – Phishing URLs add subdomains to hide the true destination by increasing the dot count
- Kim et al. (2021) – HTTPS abuse – found that HTTPS phishing campaigns now dominate phishing activity. Attackers obtain free SSL certificates to place a lock icon in the search bar.
- Alazaidah et al. (2026) – Feature selection – demonstrated that lexical URL features carry measurable predictive power, but no single feature is sufficient. Multi-feature models consistently outperform any single indicator

Overview of our Research

- **Collected URL dataset** - analyzed 630,071 cleaned URLs - (15.87% phishing / 84.13% legitimate)
- **Extracted structural features** - dots, dashes, numeric characters, URL length, path length, HTTPS wording, @ symbols
- **Conducted descriptive analysis** - identified patterns and behavioral trends in phishing URLs
- **Built predictive models** - used correlation and logistic regression to test predictive strength
- **Translated findings into real-world applications** - business security tools, employee training, and AI detection systems

Dataset Used for Research

Phishing Dataset – (Anjali, Kaggle)

- 662,591 raw URLs – 630,071 after cleaning
- 100,011 phishing URLs / 530,060 legitimate URLs
- Binary DV: Phishing = 1, Legitimate = 0

The dataset encompasses a wide array of features extracted from URLs, significantly supporting the capacity to uncover potential phishing endeavors

Key Variables

15.87% Phishing

630,071 Values

84.13% Legitimate

The dataset offers crucial insights for detecting and analyzing phishing domains embedded within URLs.

Tools Used for Research

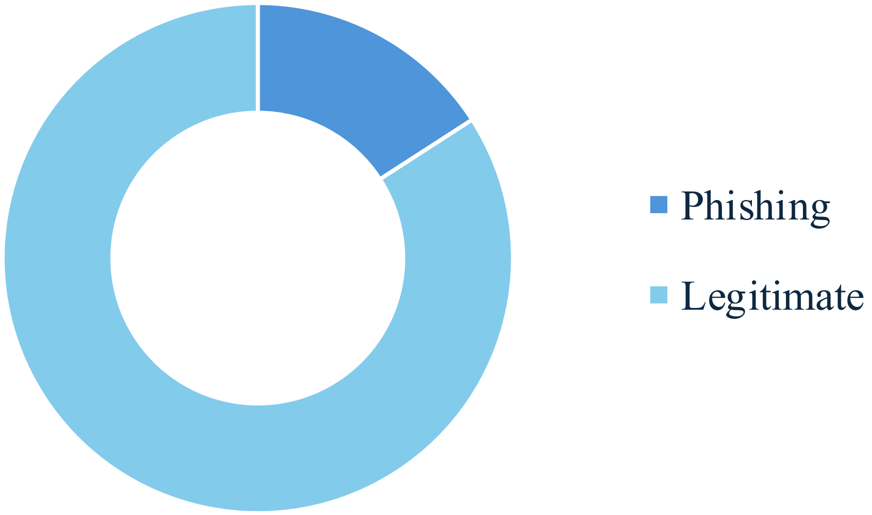
Microsoft Excel

IBM SPSS

Tableau

Python

Visual Representation



Variables Overview:

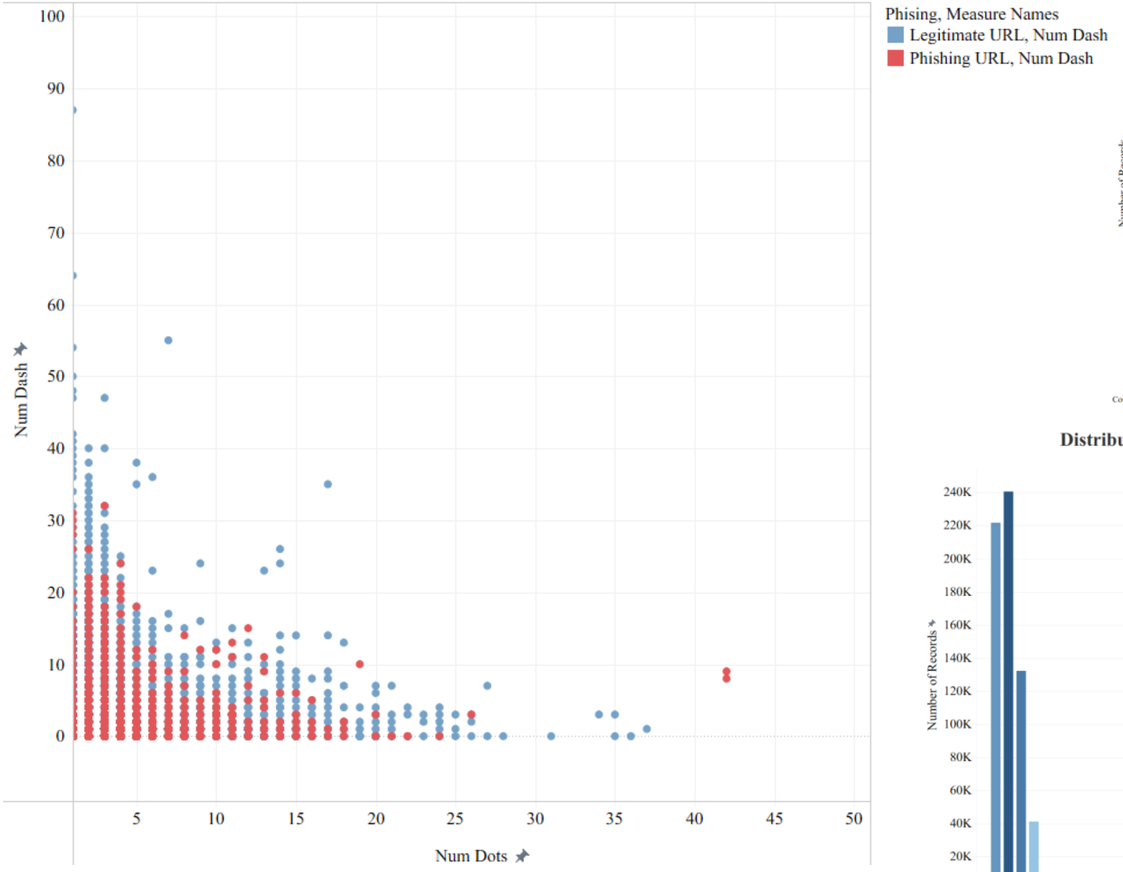
	NumDots	UrlLength	AtSymbol	NumDash	NumPercent	NumQueryComponents	IpAddress	HttpsInHostname	PathLevel	PathLength	NumNumericChars	Phising
0	3	72	0	0	0	0	0	0	5	44	0	1
1	3	144	0	0	0	2	0	0	3	16	41	1
2	3	58	0	0	0	0	0	0	2	24	0	1
3	3	79	0	1	0	0	0	0	6	50	0	1
4	3	46	0	0	0	0	0	0	4	29	2	1
5	3	42	0	1	0	0	0	0	1	12	0	1
6	2	60	0	0	0	0	0	0	5	36	1	1
7	1	30	0	0	0	0	0	0	3	11	3	1
8	8	76	0	1	0	0	0	0	2	4	2	1
9	2	46	0	0	0	0	0	0	2	25	0	1
10	5	64	0	1	0	0	0	0	2	4	3	1
11	2	47	0	0	0	0	0	0	2	18	1	1
12	2	61	0	1	0	0	0	0	2	10	0	1
13	2	35	0	0	0	0	0	0	3	12	0	1
14	2	60	0	1	0	0	0	0	2	10	0	1
15	3	73	0	0	0	0	0	0	4	33	0	1
16	3	50	0	0	0	0	1	0	5	30	10	1
17	3	59	0	1	0	0	0	0	2	16	7	1
18	2	28	0	0	0	0	0	0	3	12	1	1
19	1	59	0	0	0	0	0	0	4	40	22	1
20	1	32	0	0	0	0	0	0	4	14	4	1
21	5	52	0	0	0	0	0	0	2	20	1	1
22	2	62	0	1	0	0	0	0	6	37	0	1
23	1	105	0	2	0	0	0	0	10	87	24	1
24	4	55	0	0	0	0	0	0	2	17	0	1
25	5	134	0	3	0	0	0	0	3	56	1	1
26	2	43	0	0	0	0	0	0	3	22	0	1
27	6	210	0	2	0	5	0	0	3	54	24	1
28	2	135	0	2	0	2	0	0	7	51	1	1
29	5	85	0	0	0	0	0	0	6	69	0	1
30	2	80	0	0	0	0	0	0	3	41	17	1
31	2	44	0	0	0	0	0	0	2	18	4	1
32	3	95	0	0	4	0	0	0	9	78	16	1

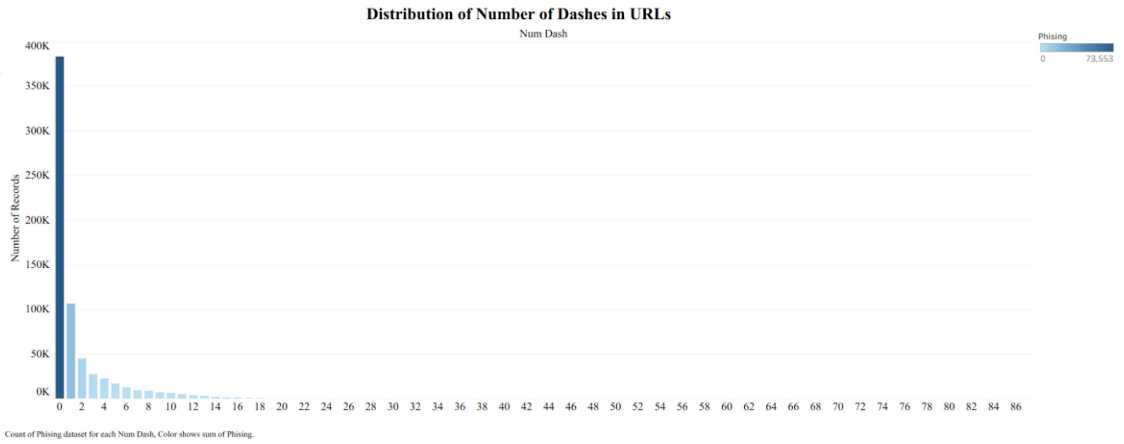
Descriptive Analysis: Number of Dots and Dashes

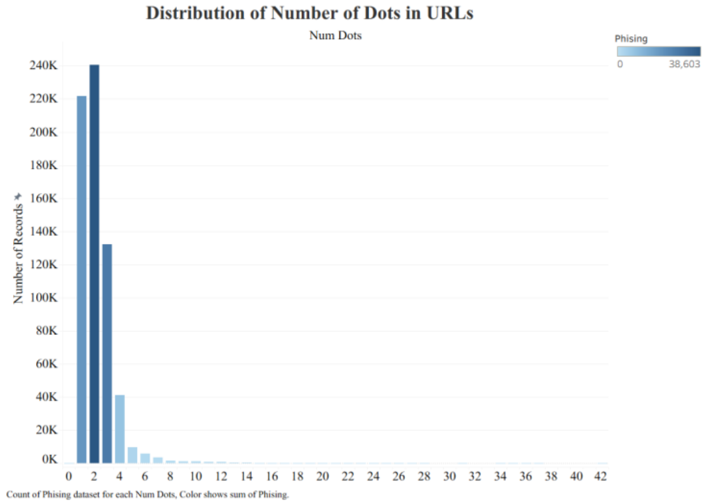
Tools Used: [Excel, Tableau]

	Mean	Median	MIN	MAX	Range	STDEV
NumDots	2.19	2	0	42	42	1.48
NumDash	1.55	0	0	87	87	2.97

ScatterPlot of URL Dots vs Dashes by Phishing Status







Kritika (2025) explains that legitimate URLs usually have about 2 dots, while phishing URLs often include extra subdomains, increasing the number of dots to hide the real domain. Attackers also use dashes to mimic trusted brand names, making the URL look more convincing.

More complex URL structures are likely to be more connected to phishing attempts. Consistent with the outcome of Descriptive analysis.

Predictive Analysis: Number of Dots and Dashes

Tools Used: [SPSS]

Correlation Analysis

NumDots vs Phishing: r = 0.098, p < .001 (more dots = more likely phishing in this dataset)
NumDash vs Phishing: r = -0.162, p < .001 (more dashes, slightly less likely phishing in this dataset)

Dots vs Dashes: r = -0.080, VIF = 1.005 for both
No multicollinearity – both included in the model

NumDots findings: Consistent with Descriptive analysis and prior research.

NumDash findings: URLs with more dashes are less likely to be classified as phishing. The dash count alone is not a strong predictor.

Correlation (r = -0.080) NumDots and NumDash are largely independent from each other. Both variables contribute unique information

Regression Analysis

R^2 .084, meaning that the independent variables together explain approximately 8.4% of the variance phishing classification. Phishing detection is complex problem that cannot be fully captured by any single pair of structural URL features alone (Zieni et al., 2023)

NumDots – Logistic Result

B = 0.140, p < .001
Exp(B) = 1.150

Each additional dot in the URL increases the odds of phishing by 15%

NumDash – Logistic Result

B = -0.355, p < .001
Exp(B) = 0.701

Each additional dash in the URL decreases the odds of phishing by 30%

Descriptive Analysis: Numeric Density & URL Length

Tools Used: [Excel, Tableau]

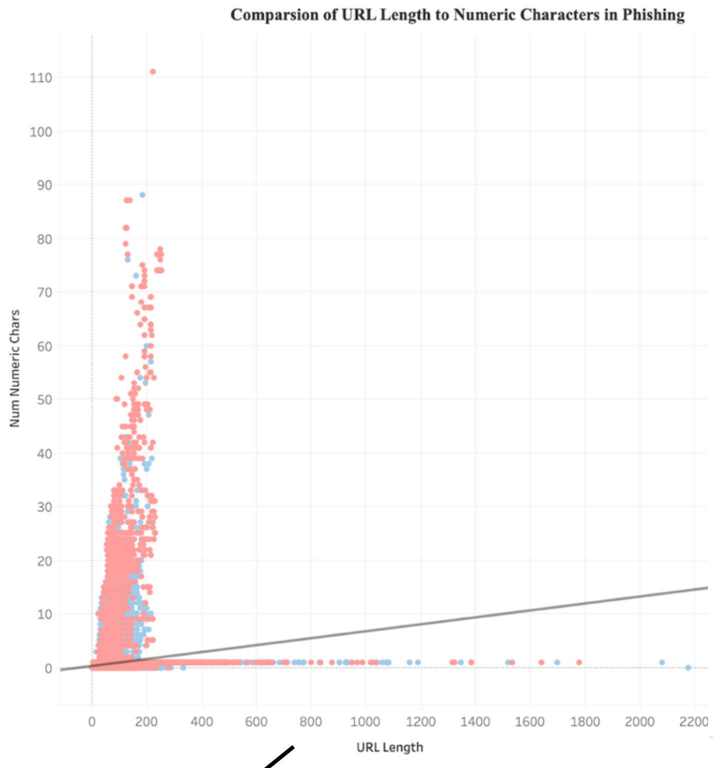
Research Question: How does numeric characters & URL length influence phishing detection?
Key Insight: Many of the URLs hold very few numbers, while some have extremely large numeric counts.
Trend: Hackers tend to incorporate high numeric usage hiding as letters in URLs.

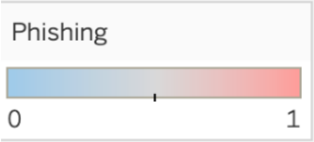
Columns: **3** (NumNumericChar + URL Length + Phishing)
IV: 2 (NumNumericChar + URL Length)
DV:1 (Phishing)

Variables	Mean	Median	Min	Max	SD
NumNumeric Char	0.65	1.00	0	111	1.43
URL Length	60.25	48.00	1	2175	44.60

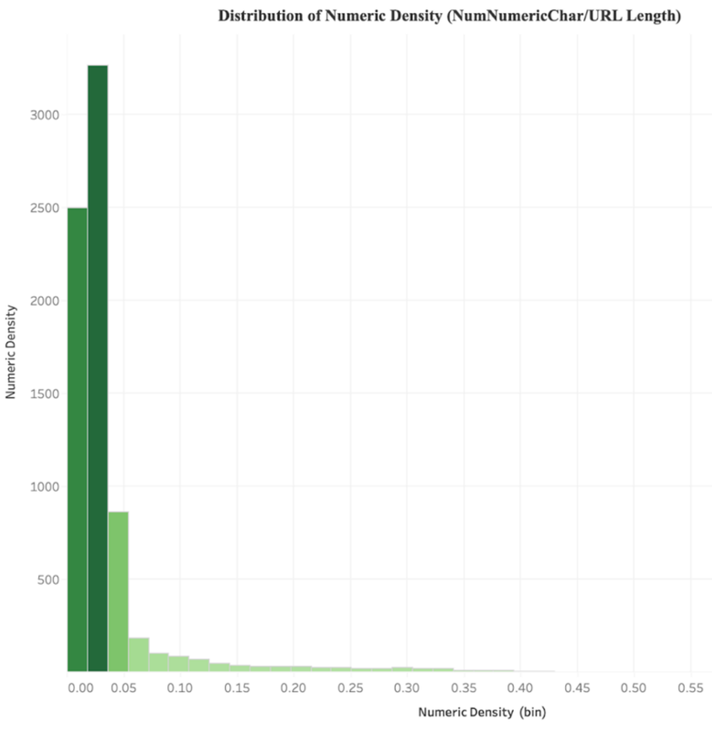
NumNumericChar:
- Left-skewed (median>mean)
- Most URLs have 0-1 numeric chars
- Counterintuitive/surprising

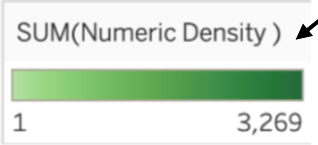
URL Length:
- Right-skewed (mean>median)
- Typical





Num Numeric Chars = 0.00647589*Url Length + 0.262342
R-Squared: 0.0407973
P-value: < 0.0001





Predictive Analysis: Numeric Density & URL Length

Tools Used: [SPSS]

Research Question: Can numeric density & URL length predict phishing activity?
Key Insight: Numeric characters and URL Length are significant, but weak indicators, neither alone are enough.
Trend: Longer URLs tend to contain more numeric characters.

Correlation Analysis → Pearson

NumNumericChars vs Phishing: **r = 0.007, p < 0.001**
URL Length vs Phishing: **r = -0.130, p < 0.001**
IVs correlated with each other: **r = 0.202**

- Both significant, but weak

Correlation alone is **NOT** predictive
→ Regression is needed

Correlations

		Phising	NumNumericChars	UrlLength
Phising	Pearson Correlation	1	.007***	-.130***
	Sig. (2-tailed)		<.001	<.001
	N	630071	630071	630071
NumNumericChars	Pearson Correlation	.007***	1	.202***
	Sig. (2-tailed)	<.001		<.001
	N	630071	662591	662591
UrlLength	Pearson Correlation	-.130***	.202***	1
	Sig. (2-tailed)	<.001	<.001	
	N	630071	662591	662591

***. Correlation at 0.001(2-tailed)

Regression Analysis → Logistic

R^2 = 0.043

- Only 4.3% of the variance
- Indicates a weak model fit, suggesting other factors to phishing detection

NumNumericChar

- **B = +0.075**
- **P = <0.001**
- **Exp(B) = 1.078**

URL Length

- **B = -0.014**
- **P = <0.001**
- **Exp(B) = 0.986**

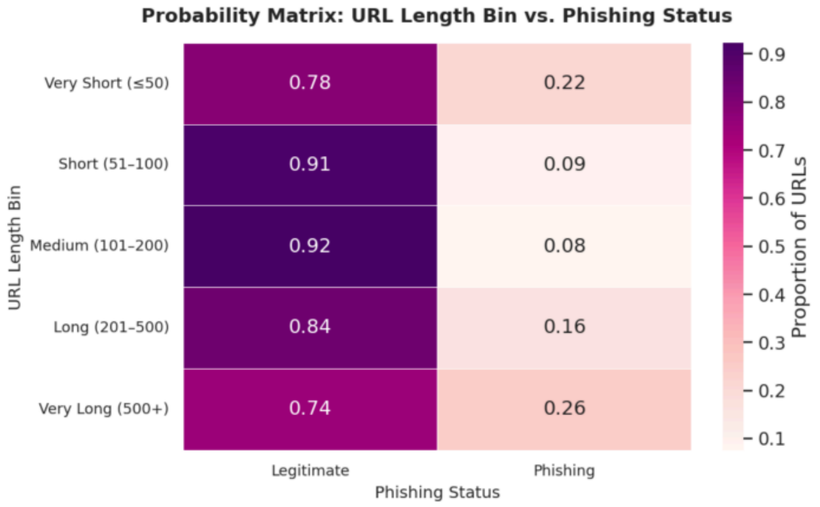
- Each additional numeric character, phishing increase by 7.8%
- Meaning that longer URLs slightly decrease the likelihood of being phishing

Descriptive Analysis: URL Length

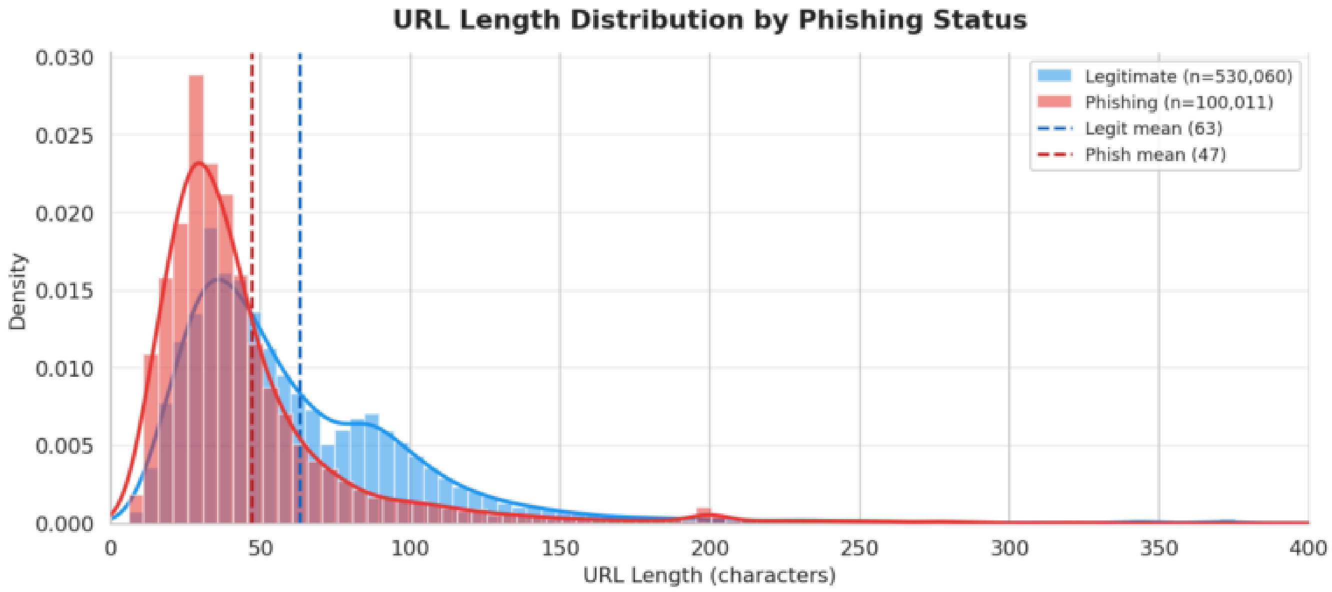
Tools Used: [Python – Pandas / Matplotlib / Seaborn]

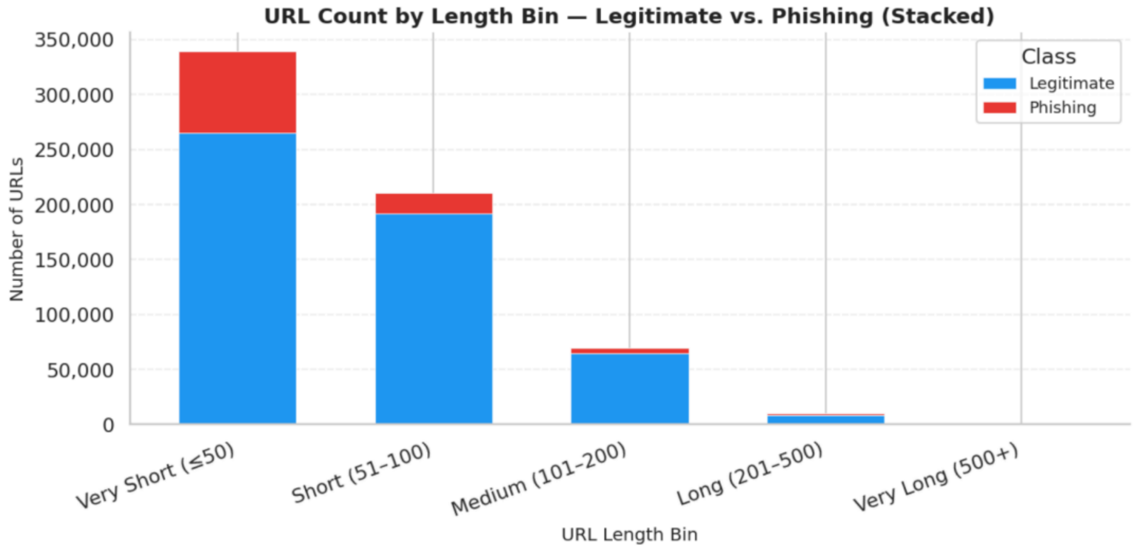
Research Question: Can URL length help us identify whether a website is phishing or legitimate?
Key Insight: Many people assume phishing URLs are long and suspicious-looking.
The data showed the opposite.
Trend:
Attackers often use shorter URLs to appear more trustworthy.

```
============================================================
        URL LENGTH vs. PHISHING — KEY FINDINGS
============================================================
  Total URLs analyzed      : 630,071
  Legitimate               : 530,060 (84.1%)
  Phishing                 : 100,011 (15.9%)

  Avg length — Legitimate : 63.0 chars
  Avg length — Phishing   : 46.9 chars
  Difference               : 16.1 chars (legit URLs are LONGER)

  Pearson correlation      : -0.1298 (negative = longer -> less phishing)
  Highest-risk bin         : Very Long (500+) -> 25.6% phishing rate

  INTERPRETATION:
  Contrary to common assumptions, phishing URLs in this
  dataset tend to be SHORTER than legitimate ones. Very
  short URLs (≤50 chars) have a ~22% phishing rate, and
  very long URLs (500+) spike again at ~26%. The safest
  range is 51–200 characters (~8–9% phishing rate).
============================================================
```



Visual Representation





Predictive Analysis: URL Length

Tools Used: [Python – Pandas / Statsmodel / SciPy]

Research Question: Can URL length by itself predict phishing?

Key Insight: URL length showed a statistically significant relationship with phishing likelihood, but it was not a strong standalone predictor.

Trend:

As URL length increases, phishing probability generally decreases, but URL length alone explains only a small portion of phishing behavior.

```
=================================================================
Intercept (B0):          -0.9694
UrlLength coef (B1):     -0.013046
p-value:                 0
Odds Ratio per +1 char:  0.9870
Odds Ratio per +50 char: 0.5208  (≈ 47.9% drop in odds)
McFadden pseudo R²:      0.0273
=================================================================
Regression equation:
  logit(P(Phising)) = -0.9694 + (-0.0130) × UrlLength
=================================================================
```

Visual Representation

```
=============================
 N = 630,071
 Phishing rate: 15.87%
==============================
Pearson r = -0.1298
p-value   = 0
N         = 630,071
==============================
```

Pearson Correlation: -0.1298

- Longer URLS = Less Phishing

```
                           Logit Regression Results
==============================================================================
Dep. Variable:                Phising   No. Observations:               441049
Model:                          Logit   Df Residuals:                   441047
Method:                           MLE   Df Model:                            1
Date:                Sun, 19 Apr 2026   Pseudo R-squ.:                 0.02735
Time:                        17:26:03   Log-Likelihood:            -1.8771e+05
converged:                       True   LL-Null:                   -1.9298e+05
Covariance Type:            nonrobust   LLR p-value:                     0.000
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const         -0.9694      0.008   -118.787      0.000      -0.985      -0.953
UrlLength     -0.0130      0.000    -89.682      0.000      -0.013      -0.013
==============================================================================
```

Descriptive Analysis: HTTPS in Hostname

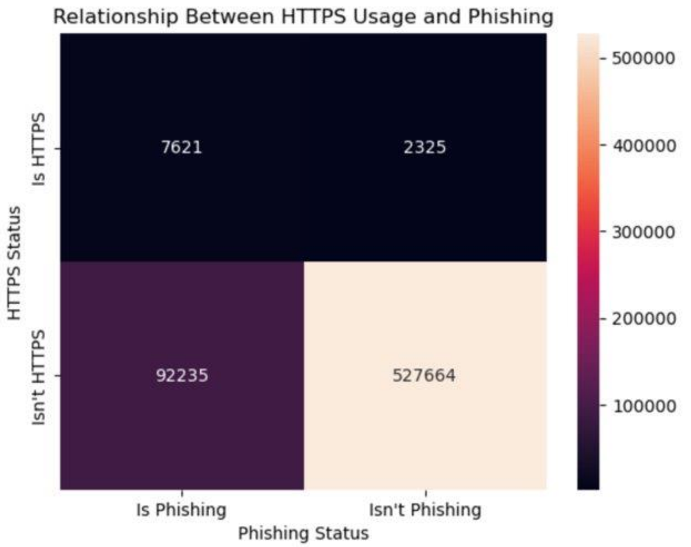
Tools Used: [Python]

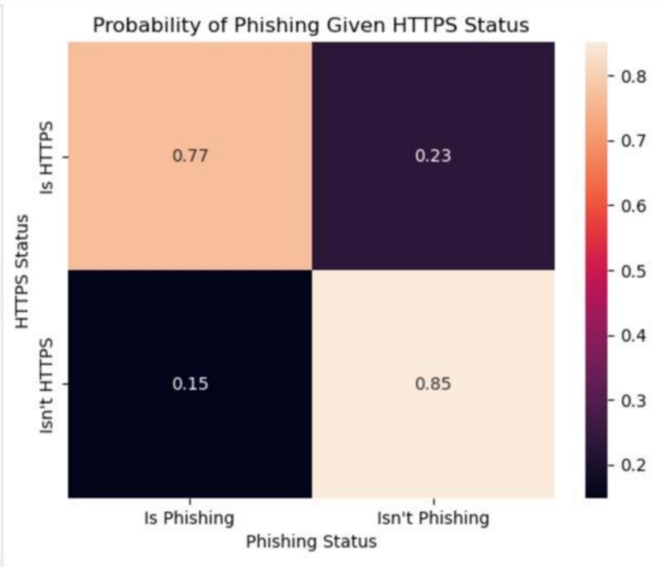
Research Question: Does including "https" inside the hostname help to identify phishing URLs?
Key Insight: HTTPS wording in the hostname was rare, but when it appeared, most of those URLs were phishing.
Trend:
Attackers can use HTTPS-related cues to make a link look safer than it actually is.

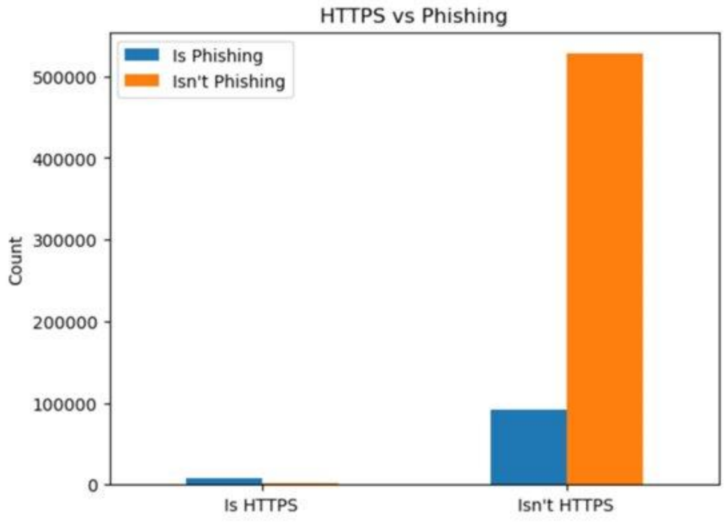
Table 1: Summary of Descriptive Statistics for Binary Variables

Variable	Mean	Median	Standard Deviation	Interpretation
HTTPS in Host Name (0 = No, 1 = Yes)	.0158	0	0.125	Only 1.58% of websites contain HTTPS in the host name, showing a highly imbalanced distribution.
Phishing (0 = No, 1 = Yes)	.1585	0	0.365	About 15.85% of websites are phishing, showing a less extreme but still uneven distribution.

Visual Representation







Predictive Analysis: HTTPS in Hostname

Tools Used: [Python]

Research Question: Can HTTPS-in-hostname predict whether a URL is phishing?
Key Insight: The model confirms the descriptive story: it is a statistically significant positive predictor of phishing.
Trend: Strong effect, but low R^2 means this signal should be layered with other URL features.

```
=== PREDICTED PROBABILITY TABLE ===
   HttpsInHostname  Predicted_Probability_of_Phishing
0                0                           0.148790
1                1                           0.764451
```

Visual Representation

```
=== LOGISTIC REGRESSION RESULTS ===
Intercept: -1.7441
Coefficient for HttpsInHostname: 2.9214
P-value for HttpsInHostname: 0.000000
Odds Ratio for HttpsInHostname: 18.5666
95% CI for Odds Ratio: (17.7260, 19.4469)
McFadden Pseudo R-squared: 0.0341

=== REGRESSION TABLE ===
          Variable  Coefficient  Std_Error  Z_Statistic  P_Value  Odds_Ratio  \
0        Intercept    -1.744120   0.003569  -488.700197      0.0    0.174799
1  HttpsInHostname     2.921362   0.023637   123.593583      0.0   18.566553

   CI_Lower_OR  CI_Upper_OR
0     0.173580     0.176026
1    17.726019    19.446944
```

Key numbers

- Pearson r = 0.2123, $p < .001$
- B = 2.9214, Odds Ratio = 18.5666
- Predicted phishing probability: 14.88% → 76.45%
- McFadden pseudo R^2 = 0.0341

Descriptive Analysis: @ Symbol & Path Length

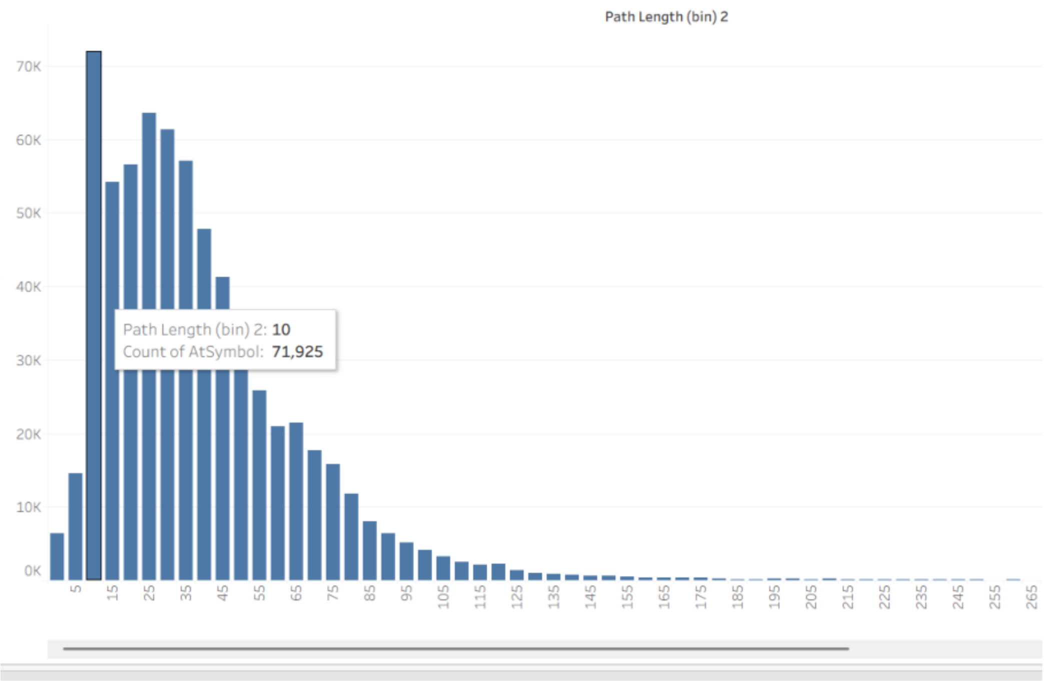
Tools Used: [Python / Tableau]

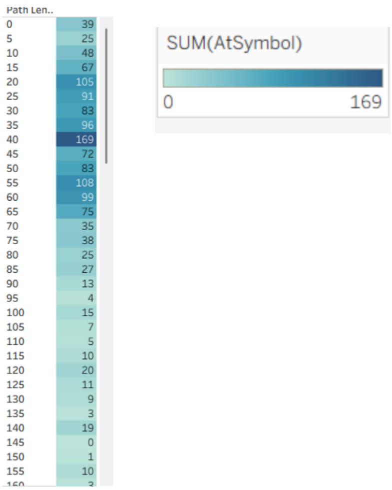
Research Question: What do @ symbol presence and URL path length reveal about suspicious URL structure?

Key Insight: @ is extremely rare, while path length varies widely across the dataset.

Trend: Most URLs have short-to-moderate paths; @ occurrences cluster around moderate path lengths rather than only very long URLs.

Visual Representation





Key Statistics

- @ mean = 0.002; median = 0; SD = 0.045.
- PathLength mean = 41.592; median = 35; SD = 29.813.
- R^2 is not used here because this slide is descriptive, not predictive.

Interpretation

- The @ symbol is meaningful descriptively because its rarity makes it stand out as suspicious.
- Path length shows real variation, but by itself it is not enough to classify a URL as phishing.
- Implication: use @ as a warning cue and path length as supporting context.

Predictive Analysis: @ Symbol & Path Length

Tools Used: [Python / Statsmodels]

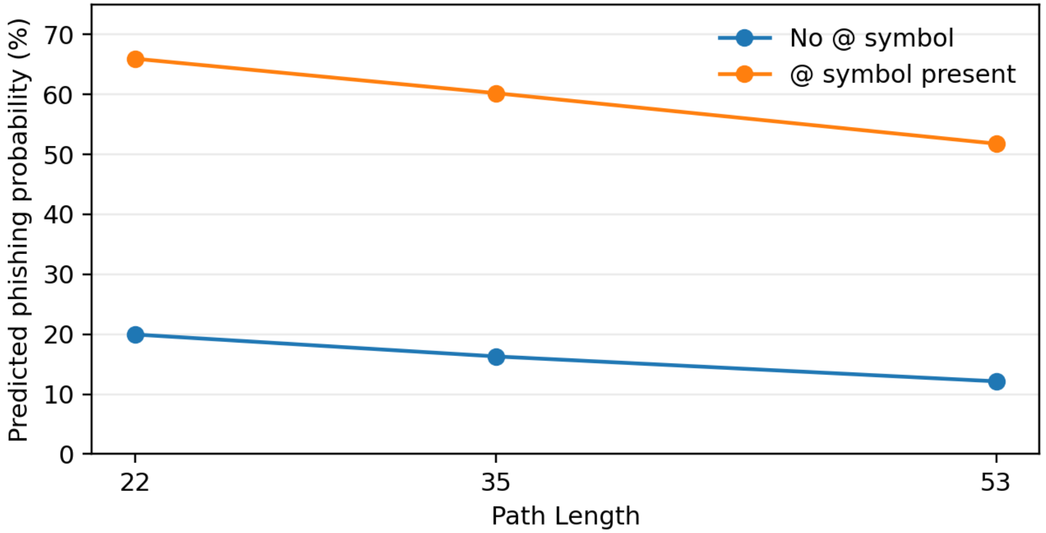
Research Question: Can @ symbol presence and path length predict whether a URL is phishing?

Key Insight: Both IVs are statistically significant, but the @ symbol is the stronger phishing predictor.

Trend: @ greatly raises predicted phishing probability; path length has a statistically significant but practically weak negative effect.

Variables: IVs = AtSymbolFlag and PathLength; DV = Phishing classification (1 = phishing, 0 = legitimate).

Visual Representation



Model Results

- AtSymbolFlag: B = 2.051, $p < .001$, OR = 7.778; strong effect size.
- PathLength: B = -0.019, $p < .001$, OR = 0.981; weak effect size.
- At path length 35, phishing rises from 16.3% without @ to 60.2% with @.

What p-values and R^2 Mean

- $p < .001$ means the relationships are statistically significant and unlikely to be random in this sample.
- However, p-values do not measure effect size or prove causation.
- McFadden pseudo-R^2 = 0.031 means weak overall model fit: these two variables explain only a small part of phishing behavior.

Interpretation / Implication

- The finding is trustworthy as a statistically significant predictive signal but limited because R^2 is low.
- Phishing detection should not rely on one rule; @ and path length should be layered with other URL features.

Comparative Summary

Feature	R^2	Exp(b)	Regression Direction	Signal Strength	Descriptive Pattern
HTTPS in Hostname	0.034	18.57	↑ Phishing	[★★★] Strongest	76.6% of HTTPS URLs are phishing (heatmap)
@ Symbol	0.031	7.78	↑ Phishing	[★★★] High-impact	<0.22% of URLs; clustered at moderate path length
NumDots	0.084	1.15	↑ Phishing	[★★] Moderate	Right-skewed; phishing at high dot counts
NumDashes	0.084	0.70	↓ Phishing	[★★] Moderate	High spread (SD=2.97); majority have zero dashes
URL Length	0.027	0.987	↓ Phishing	[★] Weak	Very short & very long both risky (U-shaped)
NumNumericChars	0.043	1.078	↑ Phishing	[★] Weak	Left-skewed; most URLs 0–1 numeric chars
Path Length	0.031	0.981	↓ Phishing	[★] Contextual	Right-skewed; moderate lengths dominate

Key Findings

Key Findings

HTTPS ≠ Safe:

- URLs with "https" in the hostname were **18.57x more likely** to be phishing

Short URLs Can Be Dangerous:

- Attackers often use shorter links to appear legitimate

Rare Features Matter:

- @ symbols increased phishing odds by **7.78x**

No Single Variable Solves Phishing:

- Highest R^2 was only **8.4%**

Practical Applications

How Businesses Can Use These Findings

Strengthen Email Security

- Automatically flag URLs containing suspici ous patterns
- Detect fake HTTPS usage before employees click

Improve Browser Protection

- Warn users about suspicious links in real-time
- Block phishing websites before credentials are entered

Train Employees

- Teach employees that HTTPS ≠ trust
- Show common phishing URL red flags

Improve AI Detection System

- Combine URL length, symbols, dots, and HTTPS signals
- Build smarter phishing detection models

Why This Matters

Phishing attacks continue evolving because attackers mimic legitimate websites.

Organizations need layered detection systems that combine automation, employee awareness, and AI-driven monitoring.

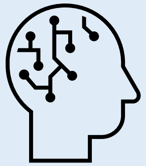
Limitations & Future Work

Limitations

- **Single Dataset**
  - Based on one Kaggle dataset
  - May not fully reflect real-world phishing traffic
- **URL-Only Analysis**
  - No email metadata
  - No webpage content analysis
  - No sender reputation signals
- **Weak Standalone Predictors**
  - Highest R^2 = **8.4%**
  - No single feature strongly predicts phishing alone

Future Work

- **Domain Intelligence**
  - Domain age
  - IP reputation scoring
- **Real-Time Detection**
  - Live phishing feeds
  - Browser monitoring tools
- **Behavioral Analytics**
  - User click behavior
  - Session monitoring
  - Device fingerprinting



Takeaway:

Future phishing defense requires real-time AI systems that combine URL analysis, behavioral monitoring, and threat intelligence.

Conclusion

Phishing Detection Requires Multiple Warning Signs:
Across both projects, our findings show that phishing URLs cannot be identified by one feature alone. The data visualization project revealed important patterns in URL behavior, while the predictive analytics project confirmed that several URL features are statistically significant predictors of phishing.

Key takeaways:

- HTTPS does **not automatically mean safe**; URLs with HTTPS in the hostname had a much higher phishing rate.
- Numeric characters, dots, dashes, URL length, path length, and the **@ symbol** all provide useful phishing clues.
- Some predictors were statistically significant but had limited predictive strength on their own.
- Outliers should not always be removed because extreme URL structures may represent real phishing behavior.
- The strongest cybersecurity approach is a **layered detection system** that combines multiple URL indicators with user awareness training.

Final message:
Phishing is adaptive and often designed to look legitimate, so users and organizations should evaluate the full URL structure instead of relying on simple assumptions like "HTTPS means safe" or "only long URLs are dangerous."

THANK YOU

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AI Disclosure:

This project used OpenAI ChatGPT (GPT-5.4) and Anthropic Claude (Opus 4.7) for outline structuring, code debugging, statistical interpretation review, and proofreading. All data analysis was conducted by the authors, and all findings, interpretations, and conclusions are the authors' own.

APPENDIX

Appendix A: Predictive Analysis of NumDots & NumDashes

Correlation

Correlations

Correlations

		NumDots_A	NumDash_A	Phising
NumDots_A	Pearson Correlation	1	-.080**	.098**
	Sig. (2-tailed)		<.001	<.001
	N	662591	662591	630071
NumDash_A	Pearson Correlation	-.080**	1	-.162**
	Sig. (2-tailed)	<.001		<.001
	N	662591	662591	630071
Phising	Pearson Correlation	.098**	-.162**	1
	Sig. (2-tailed)	<.001	<.001	
	N	630071	630071	630071

**. Correlation is significant at the 0.01 level (2-tailed).

Regression

Variables in the Equation

		B	S.E.	Wald	df	Sig.	Exp(B)
Step 0	Constant	-1.668	.003	234004.596	1	<.001	.189

Variables not in the Equation

			Score	df	Sig.
Step 0	Variables	NumDots_A	6111.549	1	<.001
		NumDash_A	16524.903	1	<.001
	Overall Statistics		21280.156	2	<.001

Block 1: Method = Enter

Omnibus Tests of Model Coefficients

		Chi-square	df	Sig.
Step 1	Step	31726.464	2	<.001
	Block	31726.464	2	<.001
	Model	31726.464	2	<.001

Multicolinearity Check

Coefficients[a]

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.	Collinearity Statistics Tolerance	Collinearity Statistics VIF
1	(Constant)	.143	.001		168.136	<.001		
	NumDash_A	-.019	.000	-.156	-125.295	<.001	.995	1.005
	NumDots_A	.021	.000	.087	70.153	<.001	.995	1.005

a. Dependent Variable: Phising

Collinearity Diagnostics[a]

Model	Dimension	Eigenvalue	Condition Index	Variance Proportions (Constant)	Variance Proportions NumDash_A	Variance Proportions NumDots_A
1	1	2.124	1.000	.06	.08	.06
	2	.711	1.728	.02	.84	.08
	3	.165	3.587	.92	.08	.86

a. Dependent Variable: Phising

Appendix A: Predictive Analysis of NumDots & NumDashes (Cont.)

Variables in the Equation

		B	S.E.	Wald	df	Sig.	Exp(B)	95% C.I.for EXP(B) Lower	Upper
Step 1[a]	NumDots_A	.140	.002	4686.373	1	<.001	1.150	1.145	1.155
	NumDash_A	-.355	.003	13776.591	1	<.001	.701	.697	.705
	Constant	-1.662	.006	73308.957	1	<.001	.190		

a. Variable(s) entered on step 1: NumDots_A, NumDash_A.

Regression

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	NumDots_A, NumDash_A[b]	.	Enter

a. Dependent Variable: Phising

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.184[a]	.034	.034	.359

a. Predictors: (Constant), NumDots_A, NumDash_A

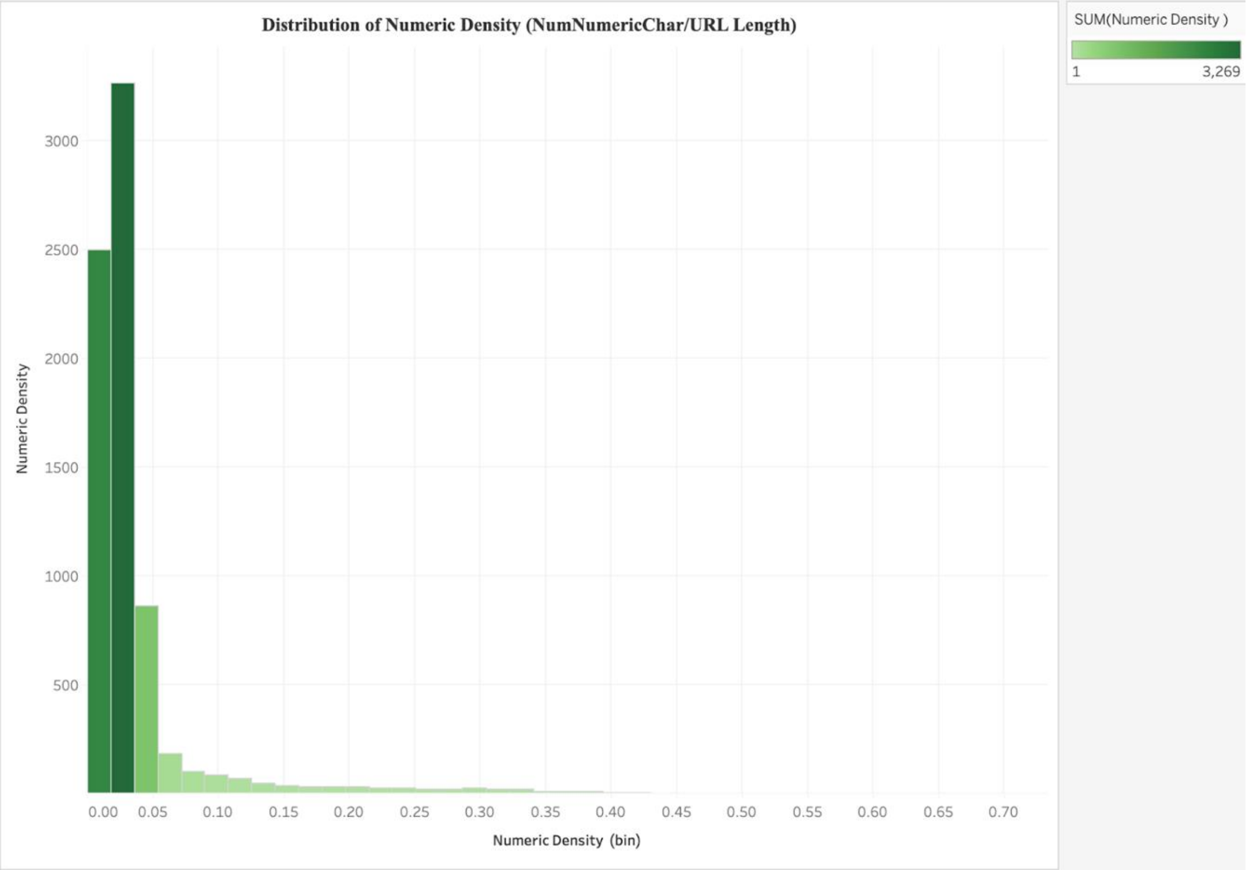
ANOVA[a]

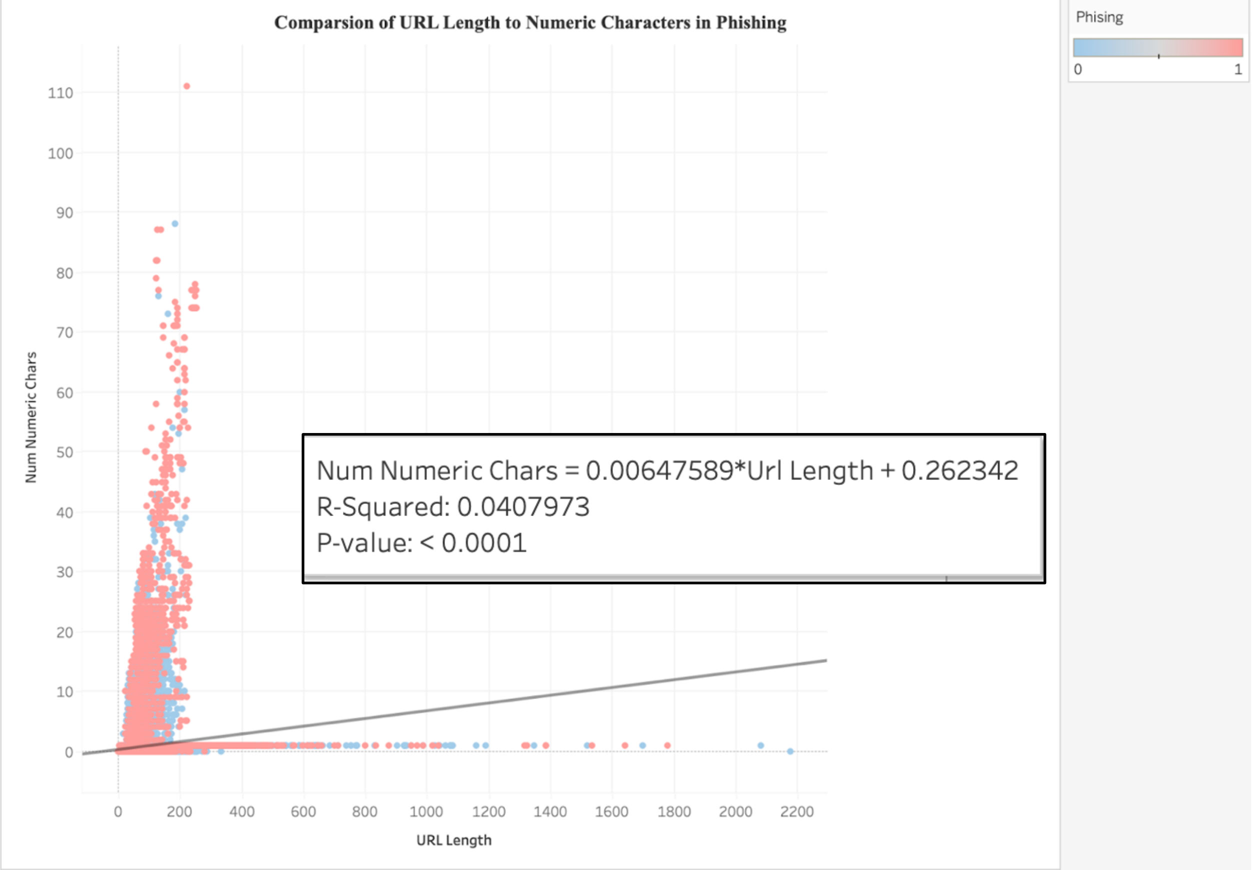
Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	2841.637	2	1420.819	11011.948	<.001[b]
	Residual	81294.644	630068	.129		
	Total	84136.281	630070			

a. Dependent Variable: Phising

b. Predictors: (Constant), NumDots_A, NumDash_A

Appendix B: Descriptive Analysis of Numeric Density & URL Length





Outlier Detection:

NumNumericChar				Key	
Calcaulte Quartiles	0	1	-->	Q1 = QUARTILE (range,1)	Q3 = QUARTILE (range,3)
IQR	1		-->	IQR = Q3 - Q1	
Outlier Threshold	-1.5	2.5	-->	LB = Q1 - 1.5*IQR	UB = Q3 + 1.5*IQR

URL Length				Key	
Calcaulte Quartiles	32	77	-->	Q1 = QUARTILE (range,1)	Q3 = QUARTILE (range,3)
IQR	45		-->	IQR = Q3 - Q1	
Outlier Threshold	-35.5	144.5	-->	LB = Q1 - 1.5*IQR	UB = Q3 + 1.5*IQR

Appendix B: Predictive Analysis of Numeric Density & URL Length

Correlations

		Phising	NumNumericChars	UrlLength
Phising	Pearson Correlation	1	.007***	-.130***
	Sig. (2-tailed)		<.001	<.001
	N	630071	630071	630071
NumNumericChars	Pearson Correlation	.007***	1	.202***
	Sig. (2-tailed)	<.001		<.001
	N	630071	662591	662591
UrlLength	Pearson Correlation	-.130***	.202***	1
	Sig. (2-tailed)	<.001	<.001	
	N	630071	662591	662591

***. Correlation at 0.001(2-tailed)

Pearson Correlations

- **Highly Positive:** *(None)*
- **Positive:** *(Phising <---> NumNumericChars), (NumNumericChars <---> UrlLength)*
- **No Linear Correlation:** *(None)*
- **Negative:** *(Phising <---> UrlLength)*
- **Highly Negative:** *(None)*

Note: Curated Help is calculated based on actual cell values, not the formatted values.

Variables not in the Equation

			Score	df	Sig.
Step 0	Variables	UrlLength	10607.596	1	<.001
		NumNumericChars	29.667	1	<.001
	Overall Statistics		11342.048	2	<.001

Appendix B: Predictive Analysis of Numeric Density & URL Length

Logistic Regression

Case Processing Summary

Unweighted Cases[a]		N	Percent
Selected Cases	Included in Analysis	630071	95.1
	Missing Cases	32520	4.9
	Total	662591	100.0
Unselected Cases		0	.0
Total		662591	100.0

a. If weight is in effect, see classification table for the total number of cases.

Dependent Variable Encoding

Original Value	Internal Value
0	0
1	1

Block 0: Beginning Block

Classification Table[a,b]

	Observed		Predicted Phising 0	Predicted Phising 1	Percentage Correct
Step 0	Phising	0	530060	0	100.0
		1	100011	0	.0
	Overall Percentage				84.1

a. Constant is included in the model.

b. The cut value is .500

Variables in the Equation

		B	S.E.	Wald	df	Sig.	Exp(B)
Step 0	Constant	-1.668	.003	234004.596	1	<.001	.189

Block 1: Method = Enter

Omnibus Tests of Model Coefficients

		Chi-square	df	Sig.
Step 1	Step	15939.629	2	<.001
	Block	15939.629	2	<.001
	Model	15939.629	2	<.001

Model Summary

Step	-2 Log likelihood	Cox & Snell R Square	Nagelkerke R Square
1	535444.912[a]	.025	.043

a. Estimation terminated at iteration number 5 because parameter estimates changed by less than .001.

Hosmer and Lemeshow Test

Step	Chi-square	df	Sig.
1	7660.644	8	<.001

Contingency Table for Hosmer and Lemeshow Test

		Phising = 0 Observed	Phising = 0 Expected	Phising = 1 Observed	Phising = 1 Expected	Total
Step 1	1	57127	59561.069	5745	3310.931	62872
	2	60354	57252.928	3055	6156.072	63409
	3	56991	54397.205	4886	7479.795	61877
	4	56939	54844.129	7119	9213.871	64058
	5	55349	53581.070	8586	10353.930	63935
	6	50190	49517.458	9896	10568.542	60086
	7	52747	53322.517	12892	12316.483	65639
	8	51172	52559.547	14536	13148.453	65708
	9	47420	49175.275	15098	13342.725	62518
	10	41771	45848.802	18198	14120.198	59969

Appendix B: Predictive Analysis of Numeric Density & URL Length

Classification Table[a]

Observed			Predicted Phising 0	Predicted Phising 1	Percentage Correct
Step 1	Phising	0	530040	20	100.0
		1	99833	178	.2
	Overall Percentage				84.2

a. The cut value is .500

Variables in the Equation

		B	S.E.	Wald	df	Sig.	Exp(B)	95% C.I.for EXP(B) Lower	95% C.I.for EXP(B) Upper
Step 1[a]	UrlLength	-.014	.000	11983.038	1	<.001	.986	.986	.986
	NumNumericChars	.075	.002	1280.062	1	<.001	1.078	1.074	1.083
	Constant	-.973	.007	20097.588	1	<.001	.378		

a. Variable(s) entered on step 1: UrlLength, NumNumericChars.

Appendix C: Descriptive Analysis of URL Length

Import:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
import matplotlib.ticker as mticker
import seaborn as sns

# Format:
sns.set_theme(style="whitegrid", font_scale=1.1)
plt.rcParams.update({
    "figure.dpi": 130,
    "axes.spines.top": False,
    "axes.spines.right": False,
    "font.family": "DejaVu Sans"
})
```

Import phishing.csv:

```
# importing phishing.csv

df = pd.read_csv("phishing.csv", index_col=0)

# Drop Rows
df = df.dropna(subset=["Phising"])
df["Phising"] = df["Phising"].astype(int)

# Columns
df["Label"] = df["Phising"].map({0: "Legitimate", 1: "Phishing"})

print(f"Dataset loaded: {len(df):,} rows × {df.shape[1]} columns")
print(f" Legitimate URLs : {(df.Phising == 0).sum():,}")
print(f" Phishing URLs   : {(df.Phising == 1).sum():,}\n")
df.head()
```

```
Dataset loaded: 630,071 rows × 13 columns
 Legitimate URLs : 530,060
 Phishing URLs   : 100,011
```

Descriptive Statistics:

```
# Step 3: Descriptive Statistics:
summary = df.groupby("Label")["UrlLength"].describe().round(2)
print("=== URL Length Summary Statistics ===")
display(summary)

legit = df[df["Phising"] == 0]["UrlLength"]
phish = df[df["Phising"] == 1]["UrlLength"]

print(f"\nMean URL Length  -->  Legitimate: {legit.mean():.1f} chars  |  Phishing: {phish.mean():.1f} chars")
print(f"Median URL Length -->  Legitimate: {legit.median():.0f} chars  |  Phishing: {phish.median():.0f} chars")
corr = df["UrlLength"].corr(df["Phising"])
print(f"Pearson Correlation (UrlLength <--> Phising): {corr:.4f}")
```

Phishing Rate by URL:

```
# Step 4: Bin URLs by length and compute phishing rate
bins   = [0, 50, 100, 200, 500, df["UrlLength"].max() + 1]
labels = ["Very Short (≤50)", "Short (51–100)", "Medium (101–200)",
          "Long (201–500)", "Very Long (500+)"]

df["LengthBin"] = pd.cut(df["UrlLength"], bins=bins, labels=labels, right=True)

bin_stats = df.groupby("LengthBin", observed=True).agg(
    Total    = ("Phising", "count"),
    Phishing = ("Phising", "sum")
)
bin_stats["Legitimate"]      = bin_stats["Total"] - bin_stats["Phishing"]
bin_stats["Phishing_Rate_%"] = (bin_stats["Phishing"] / bin_stats["Total"] * 100).round(2)

print("=== Phishing Rate by URL Length Bin ===")
display(bin_stats)
```

=== URL Length Summary Statistics ===

Label	count	mean	std	min	25%	50%	75%	max
Legitimate	530060.0	63.00	45.15	6.0	34.0	50.0	82.0	2175.0
Phishing	100011.0	46.92	43.48	1.0	26.0	36.0	52.0	1779.0

```
Mean URL Length  -->  Legitimate: 63.0 chars  |  Phishing: 46.9 chars
Median URL Length -->  Legitimate: 50 chars  |  Phishing: 36 chars
Pearson Correlation (UrlLength <--> Phising): -0.1298
```

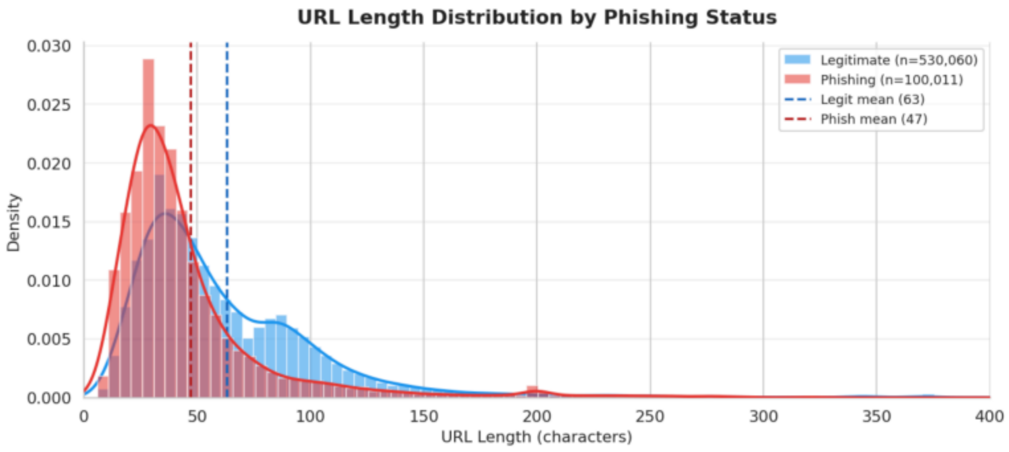
=== Phishing Rate by URL Length Bin ===

LengthBin	Total	Phishing	Legitimate	Phishing_Rate_%
Very Short (≤50)	339236	74070	265166	21.83
Short (51–100)	210380	18882	191498	8.98
Medium (101–200)	69991	5356	64635	7.65
Long (201–500)	10331	1669	8662	16.16
Very Long (500+)	133	34	99	25.56

Appendix C: Descriptive Analysis of URL Length

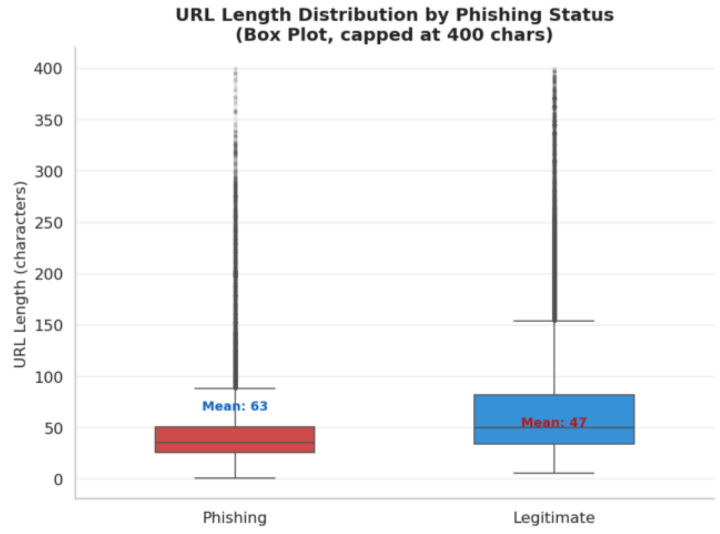
Histogram:

```python
# Plot 1: Overlapping Histogram with Kernel Density Estimation (KDE) – shows how URL lengths are distributed for each class.
# Phishing URLs are clustering at shorter lengths
CAP = 400  # cap x-axis for readability (outliers beyond 400 are rare)

fig, ax = plt.subplots(figsize=(11, 5))

for label, color, zorder in [("Legitimate", "#2196F3", 1), ("Phishing", "#E53935", 2)]:
    data = df[df["Label"] == label]["UrlLength"]
    data_capped = data[data <= CAP]
    ax.hist(data_capped, bins=80, density=True, alpha=0.55,
            color=color, label=f"{label} (n={len(data):,})", zorder=zorder)
    # Overlay KDE
    from scipy.stats import gaussian_kde
    kde = gaussian_kde(data_capped, bw_method=0.15)
    x_range = np.linspace(0, CAP, 500)
    ax.plot(x_range, kde(x_range), color=color, linewidth=2.2, zorder=zorder+1)

# Mean lines
ax.axvline(legit.mean(), color="#1565C0", linestyle="--", linewidth=1.8, label=f"Legit mean ({legit.mean():.0f})")
ax.axvline(phish.mean(), color="#B71C1C", linestyle="--", linewidth=1.8, label=f"Phish mean ({phish.mean():.0f})")

ax.set_title("URL Length Distribution by Phishing Status", fontsize=15, fontweight="bold", pad=14)
ax.set_xlabel("URL Length (characters)", fontsize=12)
ax.set_ylabel("Density", fontsize=12)
ax.set_xlim(0, CAP)
ax.legend(fontsize=10, framealpha=0.9)
ax.grid(axis="y", alpha=0.3)

plt.tight_layout()
plt.savefig("plot1_distribution.png", dpi=150, bbox_inches="tight")
plt.show()
```



Box Plot:

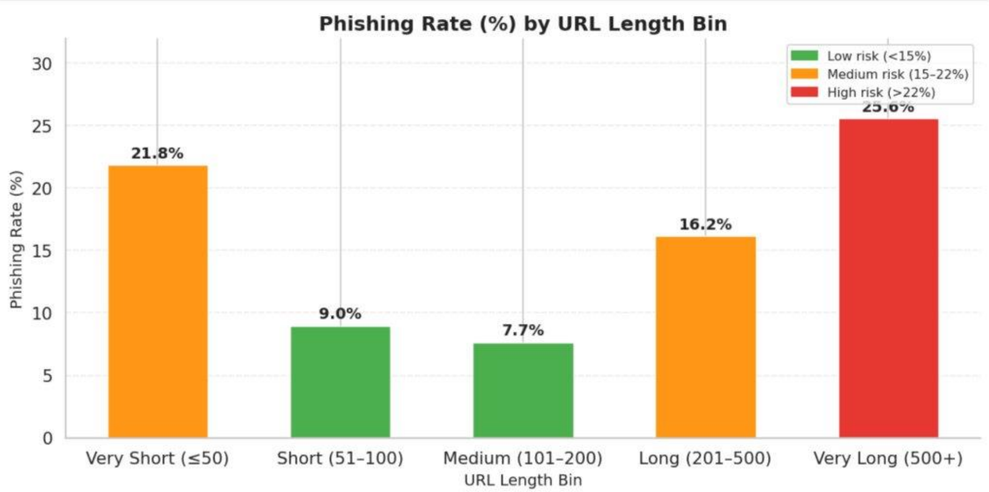
```python
# Plot 2: Box Plot – Median, Spread, and Outliers
fig, ax = plt.subplots(figsize=(8, 6))

df_cap = df[df["UrlLength"] <= 400].copy()
sns.boxplot(data=df_cap, x="Label", y="UrlLength",
            hue="Label", palette={"Legitimate": "#2196F3", "Phishing": "#E53935"},
            width=0.5, fliersize=2, flierprops=dict(alpha=0.15),
            legend=False, ax=ax)

# Annotate means
for i, (label, color) in enumerate([("Legitimate", "#1565C0"), ("Phishing", "#B71C1C")]):
    m = df[df["Label"] == label]["UrlLength"].mean()
    ax.text(i, m + 4, f"Mean: {m:.0f}", ha="center", fontsize=10,
            fontweight="bold", color=color)

ax.set_title("URL Length Distribution by Phishing Status\n(Box Plot, capped at 400 chars)",
             fontsize=14, fontweight="bold")
ax.set_xlabel("", fontsize=12)
ax.set_ylabel("URL Length (characters)", fontsize=12)
ax.grid(axis="y", alpha=0.3)

plt.tight_layout()
plt.savefig("plot2_boxplot.png", dpi=150, bbox_inches="tight")
plt.show()
```



Appendix C: Descriptive Analysis of URL Length

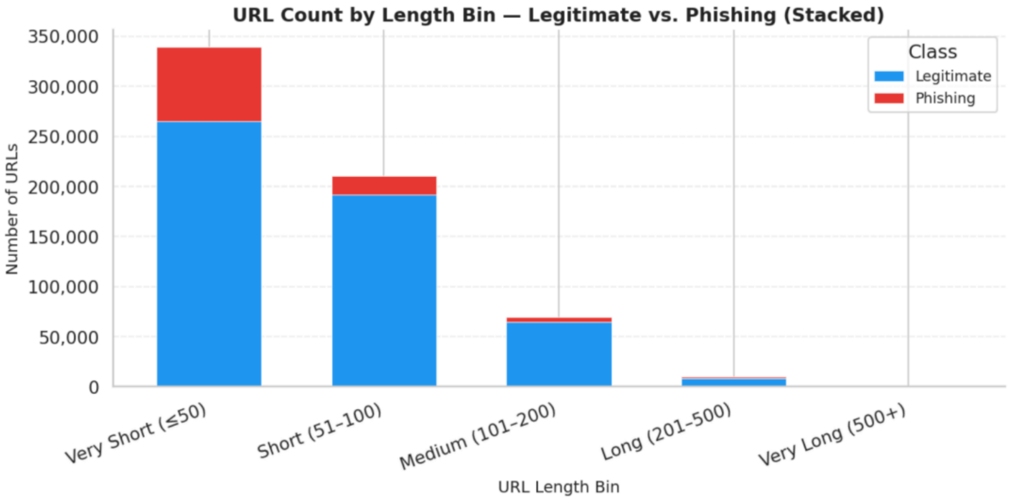
Bar Chart:

```python
# Plot 4: Phishing Rate Bar Chart – Phishing Rate by URL Length Bin – (Which URL Length ranges are most dangerous?)
rates  = bin_stats["Phishing_Rate_%"]
colors = ["#4CAF50" if r < 15 else "#FF9800" if r < 22 else "#E53935" for r in rates]

fig, ax = plt.subplots(figsize=(10, 5))
bars = ax.bar(bin_stats.index, rates, color=colors, edgecolor="white",
              linewidth=0.8, width=0.55)

# Value labels
for bar, val in zip(bars, rates):
    ax.text(bar.get_x() + bar.get_width() / 2,
            bar.get_height() + 0.3,
            f"{val:.1f}%", ha="center", va="bottom",
            fontsize=11, fontweight="bold")

ax.set_title("Phishing Rate (%) by URL Length Bin", fontsize=14, fontweight="bold")
ax.set_xlabel("URL Length Bin", fontsize=11)
ax.set_ylabel("Phishing Rate (%)", fontsize=11)
ax.set_ylim(0, rates.max() * 1.25)
ax.grid(axis="y", alpha=0.3, linestyle="--")

patches = [
    mpatches.Patch(color="#4CAF50", label="Low risk (<15%)"),
    mpatches.Patch(color="#FF9800", label="Medium risk (15–22%)"),
    mpatches.Patch(color="#E53935", label="High risk (>22%)"),
]
ax.legend(handles=patches, fontsize=9, loc="upper right")

plt.tight_layout()
plt.savefig("plot4_phishing_rate.png", dpi=150, bbox_inches="tight")
plt.show()
```



Stacked Bar Chart:

```python
# Plot 5: Stacked Bar Chart — Volume Composition per Bin
fig, ax = plt.subplots(figsize=(10, 5))

bin_counts = df.groupby(["LengthBin", "Label"], observed=True).size().unstack(fill_value=0)
bin_counts[["Legitimate", "Phishing"]].plot(
    kind="bar", stacked=True, ax=ax,
    color=["#2196F3", "#E53935"], edgecolor="white", linewidth=0.5, width=0.6
)

ax.set_title("URL Count by Length Bin — Legitimate vs. Phishing (Stacked)",
             fontsize=13, fontweight="bold")
ax.set_xlabel("URL Length Bin", fontsize=11)
ax.set_ylabel("Number of URLs", fontsize=11)
ax.set_xticklabels(ax.get_xticklabels(), rotation=20, ha="right")
ax.yaxis.set_major_formatter(mticker.FuncFormatter(lambda x, _: f"{int(x):,}"))
ax.legend(title="Class", fontsize=10)
ax.grid(axis="y", alpha=0.3, linestyle="--")

plt.tight_layout()
plt.savefig("plot5_stacked_bar.png", dpi=150, bbox_inches="tight")
plt.show()
```



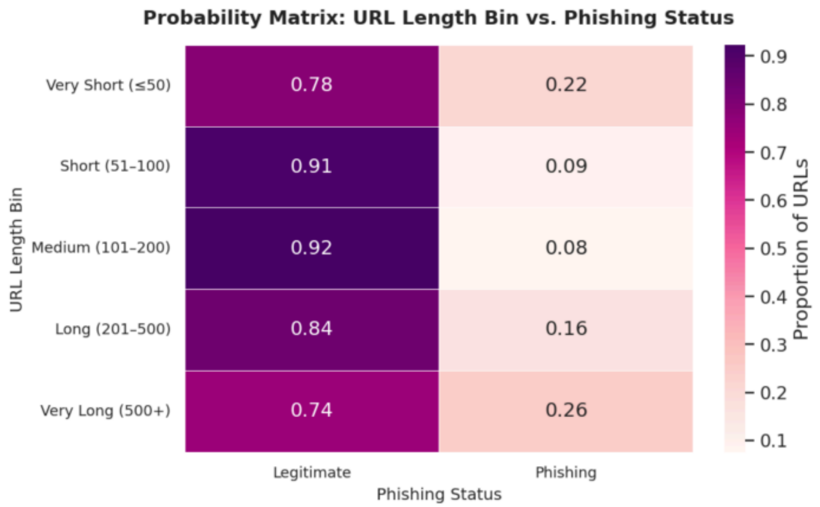
Appendix C: Descriptive Analysis of URL Length

Heat Map:

```python
# Plot 3: Probability Heatmap Matrix
pivot = df.groupby(["LengthBin", "Label"], observed=True).size().unstack(fill_value=0)
pivot_pct = pivot.div(pivot.sum(axis=1), axis=0).round(4)

fig, ax = plt.subplots(figsize=(8, 5))
sns.heatmap(pivot_pct, annot=True, fmt=".2f", cmap="RdPu",
            linewidths=0.5, linecolor="white",
            cbar_kws={"label": "Proportion of URLs"},
            ax=ax)

ax.set_title("Probability Matrix: URL Length Bin vs. Phishing Status",
             fontsize=13, fontweight="bold", pad=14)
ax.set_xlabel("Phishing Status", fontsize=11)
ax.set_ylabel("URL Length Bin", fontsize=11)
ax.set_xticklabels(["Legitimate", "Phishing"], fontsize=10)
ax.set_yticklabels(ax.get_yticklabels(), rotation=0, fontsize=10)

plt.tight_layout()
plt.savefig("plot3_heatmap.png", dpi=150, bbox_inches="tight")
plt.show()
```

Interpretation:

```python
# Key Findings & Interpretation:

top_bin  = bin_stats["Phishing_Rate_%"].idxmax()
top_rate = bin_stats["Phishing_Rate_%"].max()

print("=" * 60)
print("         URL LENGTH vs. PHISHING — KEY FINDINGS")
print("=" * 60)
print(f"  Total URLs analyzed     : {len(df):,}")
print(f"  Legitimate              : {(df.Phising==0).sum():,} ({(df.Phising==0).mean()*100:.1f}%)")
print(f"  Phishing                : {(df.Phising==1).sum():,} ({(df.Phising==1).mean()*100:.1f}%)")
print()
print(f"  Avg length — Legitimate : {legit.mean():.1f} chars")
print(f"  Avg length — Phishing   : {phish.mean():.1f} chars")
print(f"  Difference              : {legit.mean()-phish.mean():.1f} chars (legit URLs are LONGER)")
print()
print(f"  Pearson correlation     : {df['UrlLength'].corr(df['Phising']):.4f} (negative = longer -> less phishing)")
print(f"  Highest-risk bin        : {top_bin} -> {top_rate:.1f}% phishing rate")
print()
print("  INTERPRETATION:")
print("  Contrary to common assumptions, phishing URLs in this")
print("  dataset tend to be SHORTER than legitimate ones. Very")
print("  short URLs (≤50 chars) have a ~22% phishing rate, and")
print("  very long URLs (500+) spike again at ~26%. The safest")
print("  range is 51–200 characters (~8–9% phishing rate).")
print("=" * 60)
```




```
============================================================
         URL LENGTH vs. PHISHING — KEY FINDINGS
============================================================
  Total URLs analyzed     : 630,071
  Legitimate              : 530,060 (84.1%)
  Phishing                : 100,011 (15.9%)

  Avg length — Legitimate : 63.0 chars
  Avg length — Phishing   : 46.9 chars
  Difference              : 16.1 chars (legit URLs are LONGER)

  Pearson correlation     : -0.1298 (negative = longer -> less phishing)
  Highest-risk bin        : Very Long (500+) -> 25.6% phishing rate

  INTERPRETATION:
  Contrary to common assumptions, phishing URLs in this
  dataset tend to be SHORTER than legitimate ones. Very
  short URLs (≤50 chars) have a ~22% phishing rate, and
  very long URLs (500+) spike again at ~26%. The safest
  range is 51–200 characters (~8–9% phishing rate).
============================================================
```

Appendix C: Predictive Analysis of URL Length

Import:

```
# Import

import pandas as pd
import numpy as np
from scipy import stats
import statsmodels.api as sm
from sklearn.model_selection import train_test_split
from sklearn.metrics import roc_auc_score, roc_curve, accuracy_score

CSV_PATH = "phishing.csv"
```

Read Data Set and Organize

```
# Read Data Set & Organize
df = pd.read_csv("phishing.csv")                                    # read the dataset into a DataFrame
df = df.drop(columns=[c for c in df.columns if c.startswith("Unnamed")])  # drop the unnamed index column if present
df = df.dropna(subset=["Phising"]).reset_index(drop=True) # remove rows where the target label is missing
df["Phising"] = df["Phising"].astype(int)                           # cast target from float (0.0/1.0) to int (0/1)
print('='*30)
print(f"N = {len(df):,}")                                            # total number of rows after cleaning
print(f"Phishing rate: {df['Phising'].mean()*100:.2f}%")             # % of rows labeled phishing (class balance)
print('='*30)
df[["UrlLength", "Phising"]].head()                                  # preview first 5 rows of the two columns we'll use
```

Train & Testing Data

```
# Train & Testing Data

X = df[["UrlLength"]]                               # independent variable (as a DataFrame, not Series)
y = df["Phising"]                                   # dependent variable (the label we're predicting)

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.30,                                 # hold out 30% of the data for testing
    random_state=42,                                # fixed seed so results are reproducible
    stratify=y                                      # keep the 15.87% phishing rate in both train & test
)
print('='*50)
print(f"Train: {len(X_train):,} rows  ({y_train.mean()*100:.2f}% phishing)")
print(f"Test:  {len(X_test):,} rows  ({y_test.mean()*100:.2f}% phishing)")
print('='*50)
```

Threshold Tuning

```
# Threshold Tuning

fpr, tpr, thresholds = roc_curve(y_test, test_probs)      # arrays of FPR, TPR, and cutoff values along the ROC curve
best_idx = np.argmax(tpr - fpr)                           # Youden's J picks the threshold maximizing TPR − FPR
best_thr = thresholds[best_idx]                           # the corresponding probability cutoff
print('='*50)
print(f"Optimal threshold (Youden's J): {best_thr:.4f}")
print(f"  True Positive Rate  (recall): {tpr[best_idx]:.4f}")    # % of actual phishing URLs correctly flagged
print(f"  False Positive Rate:          {fpr[best_idx]:.4f}")    # % of legit URLs incorrectly flagged
print('='*50)
preds_tuned = (test_probs >= best_thr).astype(int)        # reclassify using the tuned threshold
print(f"At threshold = {best_thr:.4f}:")
print(f"  Accuracy: {accuracy_score(y_test, preds_tuned):.4f}")
print(f"  Phishing predictions: {preds_tuned.sum():,}")
print('='*50)
```

Training & Testing

```
==================================================
Train: 441,049 rows  (15.87% phishing)
Test:  189,022 rows  (15.87% phishing)
==================================================
```

Threshold Tuning

```
==================================================
Optimal threshold (Youden's J): 0.1760
  True Positive Rate  (recall): 0.6692
  False Positive Rate:          0.4194
==================================================
At threshold = 0.1760:
  Accuracy: 0.5947
  Phishing predictions: 86,767
==================================================
```

Appendix C: Predictive Analysis of URL Length

Logistic Regression

```
# Logistic Regression

model = sm.Logit(                                    # Logit = logistic regression
    y_train,                                         # dependent variable (training)
    sm.add_constant(X_train)                         # adds the intercept column (all 1s) to the IVs
).fit(disp=False)                                    # fit the model; disp=False hides iteration output
print(model.summary())                               # full regression output: coef, SE, z, p, CI, pseudo R², etc.
```

Key Regression Numbers

```
# Key Regression Numbers

b0 = model.params["const"]                           # intercept coefficient
b1 = model.params["UrlLength"]                       # slope coefficient for UrlLength
or_1  = np.exp(b1)                                   # odds ratio per +1 character (exp of the coefficient)
or_50 = np.exp(b1 * 50)                              # odds ratio per +50 characters (easier to interpret)

print('='*70)
print(f"Intercept (B0):           {b0:.4f}")
print(f"UrlLength coef (B1):      {b1:.6f}")
print(f"p-value:                  {model.pvalues['UrlLength']:.3g}")  # significance of B1
print(f"Odds Ratio per +1 char:   {or_1:.4f}")
print(f"Odds Ratio per +50 char:  {or_50:.4f}  (≈ {(1-or_50)*100:.1f}% drop in odds)")
print(f"McFadden pseudo R²:       {model.prsquared:.4f}") # % of variance in log-odds explained (0–1 scale)
print('='*70)
print(f"Regression equation:")
print(f"  logit(P(Phising)) = {b0:.4f} + ({b1:.4f}) × UrlLength")  # the fitted equation
print('='*70)
```

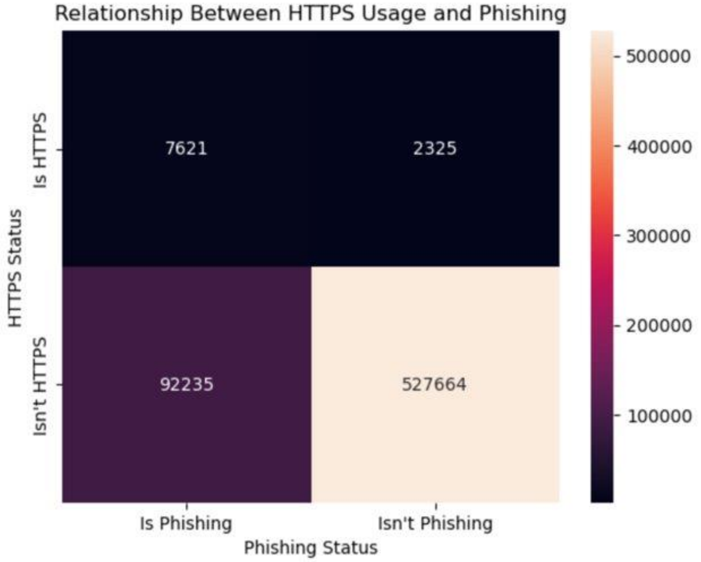
Logistic Regression

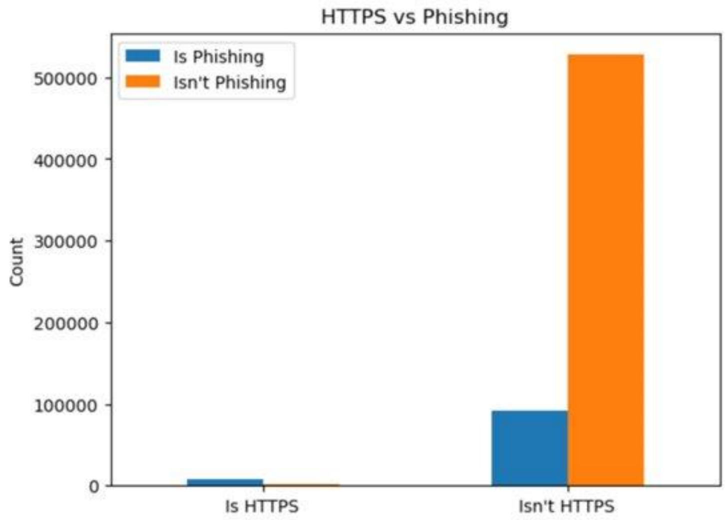
```
                           Logit Regression Results
==============================================================================
Dep. Variable:                Phising   No. Observations:               441049
Model:                          Logit   Df Residuals:                   441047
Method:                           MLE   Df Model:                            1
Date:                Sun, 19 Apr 2026   Pseudo R-squ.:                 0.02735
Time:                        17:26:03   Log-Likelihood:            -1.8771e+05
converged:                       True   LL-Null:                   -1.9298e+05
Covariance Type:            nonrobust   LLR p-value:                     0.000
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const         -0.9694      0.008   -118.787      0.000      -0.985      -0.953
UrlLength     -0.0130      0.000    -89.682      0.000      -0.013      -0.013
==============================================================================
```

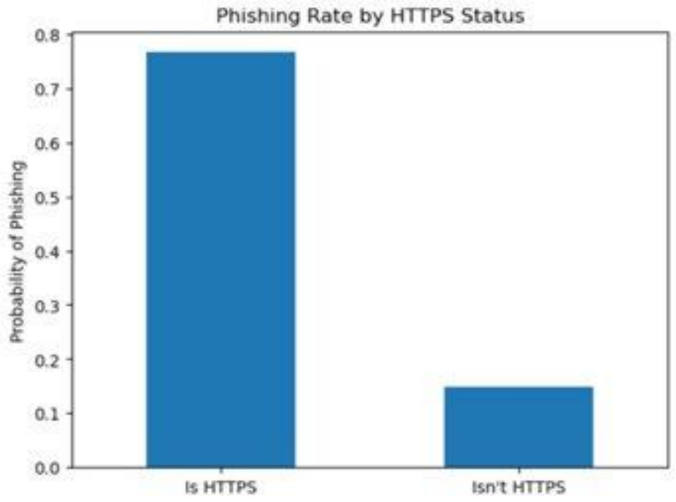
Key Regression Numbers

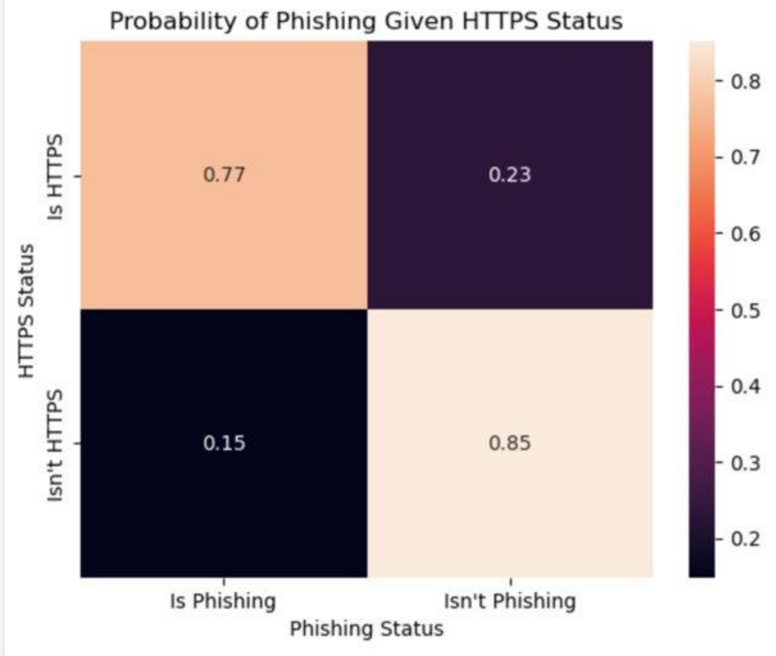
```
======================================================================
Intercept (B0):           -0.9694
UrlLength coef (B1):      -0.013046
p-value:                  0
Odds Ratio per +1 char:   0.9870
Odds Ratio per +50 char:  0.5208  (≈ 47.9% drop in odds)
McFadden pseudo R²:       0.0273
======================================================================
Regression equation:
  logit(P(Phising)) = -0.9694 + (-0.0130) × UrlLength
======================================================================
```

Appendix D: Descriptive Analysis of HTTPS in Hostname









Main descriptive takeaway: HTTPS-in-hostname was uncommon in the dataset, but when present it was heavily concentrated in phishing URLs.

Counts: 619,899 URLs without HTTPS in hostname vs. 10,172 URLs with it.
Rates: 14.88% phishing without HTTPS in hostname vs. 76.45% phishing with it.

Appendix D: Predictive Analysis of HTTPS in Hostname

```
=== CORRELATION ANALYSIS ===
Pearson correlation (phi coefficient for binary variables): 0.2123
P-value: 0.000000
```

```
=== PREDICTED PROBABILITY TABLE ===
   HttpsInHostname  Predicted_Probability_of_Phishing
0                0                           0.148790
1                1                           0.764451
```

```
=== LOGISTIC REGRESSION RESULTS ===
Intercept: -1.7441
Coefficient for HttpsInHostname: 2.9214
P-value for HttpsInHostname: 0.000000
Odds Ratio for HttpsInHostname: 18.5666
95% CI for Odds Ratio: (17.7260, 19.4469)
McFadden Pseudo R-squared: 0.0341

=== REGRESSION TABLE ===
          Variable  Coefficient  Std_Error  Z_Statistic  P_Value  Odds_Ratio  \
0        Intercept    -1.744120   0.003569  -488.700197      0.0    0.174799
1  HttpsInHostname     2.921362   0.023637   123.593583      0.0   18.566553

   CI_Lower_OR  CI_Upper_OR
0     0.173580     0.176026
1    17.726019    19.446944
```

Key numbers to cite on the presentation slide:

- Pearson $r = 0.2123$, $p < .001$
- $B = 2.9214$, Odds Ratio = 18.5666
- Predicted phishing probability: 14.88% → 76.45%
- McFadden pseudo $R^2 = 0.0341$

Appendix E: Descriptive Analysis of The @ Symbol & Path Length

Dataset / source: Simaanjali TES Upload Dataset (Kaggle); N = 630,070 cleaned URLs

Variable	Mean	Median	Std. Dev.
AtSymbolFlag	0.002074	0	0.045498
PathLength	41.591814	35	29.813151
Phishing	0.158728	0	0.365

- @ symbol is absent from the typical URL, making it a rare warning cue when present.
- Path length is right-skewed: most URLs are short-to-moderate, with fewer very long paths.
- Descriptive work has no R^2 because it summarizes distributions rather than estimating a prediction model.

What this appendix evidence supports

- Phishing URLs may appear normal in length while containing subtle suspicious features.
- The @ symbol should be treated as a high-attention cue, not as a common URL trait.

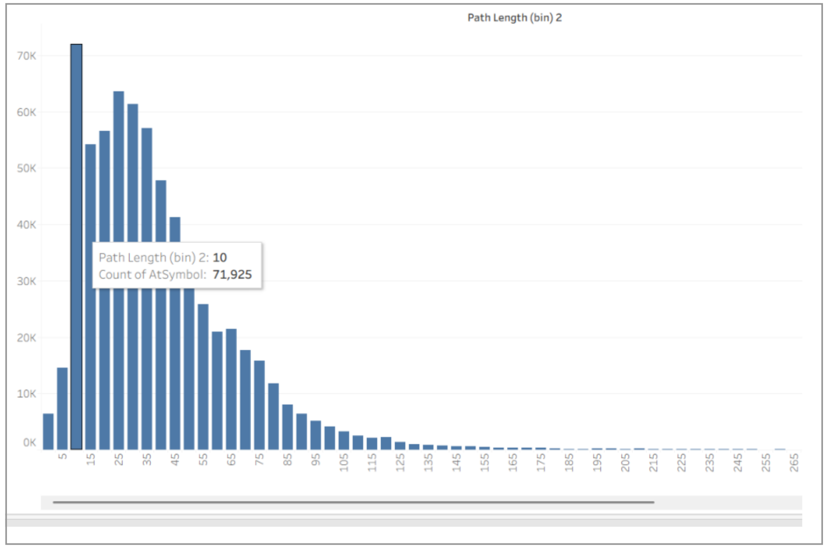


Figure E1. Histogram of PathLength showing right-skew and concentration in lower/middle bins.

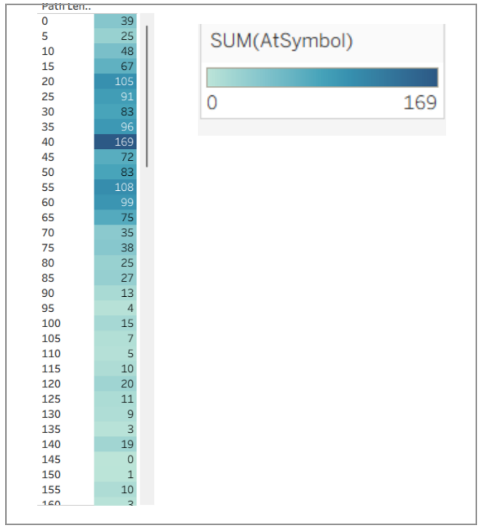


Figure E2. Heatmap showing @ counts concentrated near moderate path lengths.

Presenter note: use this slide briefly to show the descriptive visuals and the statistical summary behind the main slide.

Appendix E: Descriptive Evidence — Python Code and Output

Purpose: show the actual Python workflow used to clean variables, recode AtSymbol into AtSymbolFlag, and calculate descriptive statistics/correlations.

```python
import pandas as pd
import numpy as np
from scipy.stats import pearsonr, norm
from scipy.optimize import minimize

# ========================
# 1. LOAD DATA
# ========================
file_path = "Phising_dataset_predict (2).xlsx"
df = pd.read_excel(file_path)

# Keep only the variables needed for this story
df = df[["AtSymbol", "PathLength", "Phising"]].copy()

# Drop missing values
df = df.dropna(subset=["AtSymbol", "PathLength", "Phising"])

# Recode AtSymbol into a binary flag
# If AtSymbol is greater than 0, code as 1; otherwise 0
df["AtSymbolFlag"] = df["AtSymbol"].apply(lambda x: 1 if x > 0 else 0)

# Make sure variables are numeric
df["PathLength"] = pd.to_numeric(df["PathLength"], errors="coerce")
df["Phising"] = pd.to_numeric(df["Phising"], errors="coerce").astype(int)

# Drop any rows that became invalid after conversion
df = df.dropna(subset=["PathLength", "Phising"])

print("=== DATA PREVIEW ===")
print(df[["AtSymbolFlag", "PathLength", "Phising"]].head())
print("\n=== SAMPLE SIZE ===")
print(len(df))
```

Figure E3. Python preparation code: loading data, keeping AtSymbol/PathLength/Phishing, dropping missing values, and recoding @ into a binary flag.

```
=== DESCRIPTIVE STATISTICS ===
N = 630,070

AtSymbolFlag: mean = 0.002074, median = 0,
std. dev. = 0.045498

PathLength: mean = 41.591814, median = 35,
std. dev. = 29.813151

Phising: mean = 0.158728

=== CORRELATION MATRIX ===
              AtSymbolFlag  PathLength   Phising
AtSymbolFlag      1.000000    0.017176  0.045873
PathLength        0.017176    1.000000 -0.138135
Phising           0.045873   -0.138135  1.000000

Pearson r (AtSymbolFlag, Phising) = 0.045873, p = 1.352e-290
Pearson r (PathLength, Phising) = -0.138135, p = 0.000e+00
Pearson r (AtSymbolFlag, PathLength) = 0.017176, p = 2.488e-42
```

Figure E4. Python output: descriptive statistics, correlation matrix, and Pearson correlations.

Key output values to report

- AtSymbolFlag vs Phishing: $r = 0.045873$, $p \approx 1.352\text{e-}290$; statistically significant but weak.
- PathLength vs Phishing: $r = -0.138135$, $p < .001$; statistically significant but still weak.
- AtSymbolFlag vs PathLength: $r = 0.017176$, so multicollinearity is not a major issue.

Appendix E: Predictive Analysis of The @ Symbol & Path Length

Dependent variable: Phishing (1 = phishing, 0 = legitimate). Independent variables: AtSymbolFlag and PathLength. Model: binary logistic regression.

Variable	B	p-value	Odds Ratio	Effect
AtSymbolFlag	2.051	$< .001$	7.778	Strong
PathLength	-0.019	$< .001$	0.981	Weak
Model fit	—	—	Pseudo-R^2 = 0.031	Low

- p-values below .001 mean the predictors are statistically trustworthy in this dataset.
- Pseudo-R^2 = 0.031 means the model has weak overall explanatory power and should not be used alone.
- Practical implication: @ is a strong red flag, while path length is a supporting contextual cue.

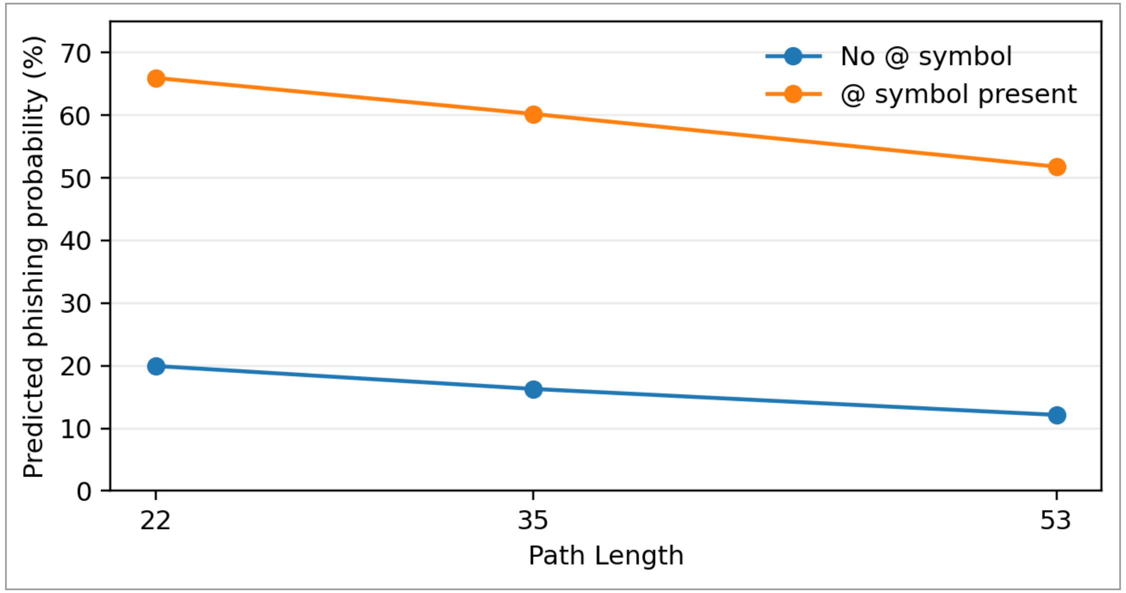


Figure E5. Predicted phishing probabilities at representative path lengths, with and without @.

Main predictive takeaway

- At path length 35, predicted phishing rises from 16.3% without @ to 60.2% with @.
- The trend for @ is meaningful; the path length trend is statistically significant but practically small.

Presenter note this is the slide to use if the professor asks where the regression and R^2 values are shown.

Appendix E: Predictive Evidence — Regression Output and Probabilities

This slide documents the actual model output so the presentation lists the regression work behind the main findings.

```
=== LOGISTIC REGRESSION RESULTS ===
McFadden pseudo R-squared = 0.030610

Variable        Coef.       Std.Err.        z         p-value    Odds Ratio
const        -0.970370    0.006589   -147.279   0.000e+00    0.378943
AtSymbolFlag 2.051362    0.057935    35.408   1.289e-274    7.778488
PathLength   -0.019043    0.000168  -113.169   0.000e+00    0.981138

95% CI for AtSymbolFlag odds ratio: 6.943537 to 8.713841
95% CI for PathLength odds ratio: 0.980814 to 0.981461

=== PREDICTED PROBABILITIES ===
AtSymbolFlag = 0, PathLength = 22, probability = 0.199519
AtSymbolFlag = 1, PathLength = 22, probability = 0.659722
AtSymbolFlag = 0, PathLength = 35, probability = 0.162893
AtSymbolFlag = 1, PathLength = 35, probability = 0.602167
AtSymbolFlag = 0, PathLength = 53, probability = 0.121359
AtSymbolFlag = 1, PathLength = 53, probability = 0.517927
```

Figure E6. Logistic regression output: coefficients, standard errors, z-values, p-values, odds ratios, confidence intervals, and predicted probabilities.

How to explain the p-values and R^2

- $p < .001$: the results are unlikely to be random, so the association is statistically reliable in this sample.
- Pseudo-$R^2 = 0.031$: the model explains only a small amount of the phishing classification pattern.
- Trustworthy ≠ complete: the predictors matter, but many other URL/domain features are still needed.

Overall implication: phishing detection should combine rare lexical cues, structural URL features, and other indicators in a layered model.